

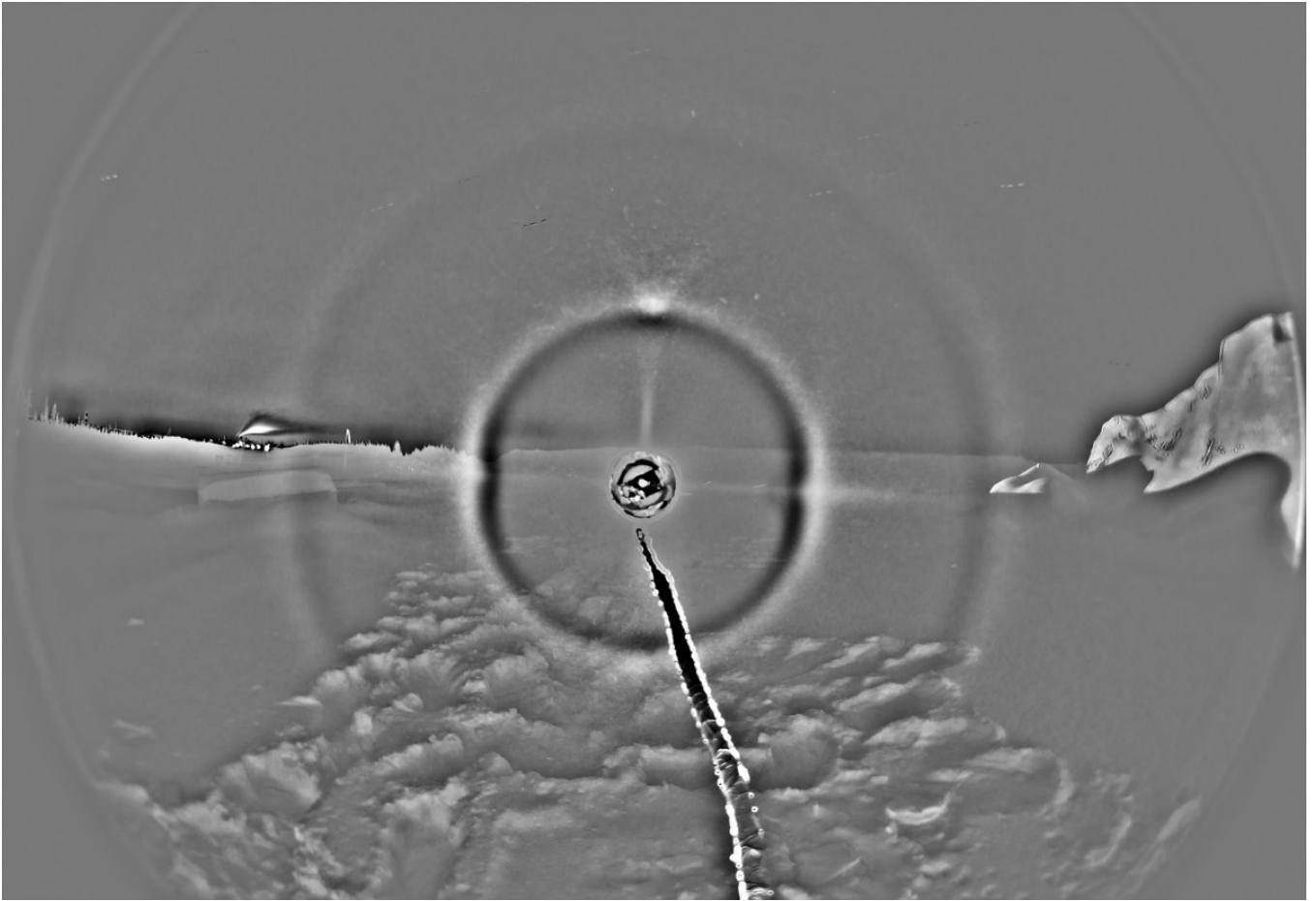
Thread - 2024-04-02

<https://x.com/RiikonenMarko/status/1775191294958027214>

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5/6 February 2024 Rovaniemi, Anttilanvaara. In this thread are three differently initiated ice fog displays: from snow that I threw in the air (first three photos), from an unknown source (fourth photo) and from Suosiola plume (next thread). The snow-throwing born displays were perhaps the most striking I have seen of the type, but they were still the same old puffed combo of tangent arc, parhelia, 22 and 46° halo (both stacks have around 10-15 frames, 30s exp). The unknown-source display was the night's second similar display (6 frames, 30s exp), I didn't get the first photographed. They were short-lived and had also subanthelic arc visible to the eye. My guess is they spawned from unloading of trucks at the snow depo 700 meters away. Given the constant traffic, I might have had a more productive night had I been there instead (but you have to make 7 km detour to reach that location as there is no more crossing over the railway that runs in between).







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5/6 February 2024 Rovaniemi, Anttilanvaara. The Suosiola plume display. Since I started in the evening I had watched Suosiola plant plume depositing constantly ice fog nearby in the industrial area. Because of light pollution, there was no sense going there. Hour by hour the plume shifted ever so slightly towards Anttilanvaara until near the morning it unhurriedly swept over allowing me to take 111 frames with 15s exposures. The display is not much to go by, but crystals make it more interesting.

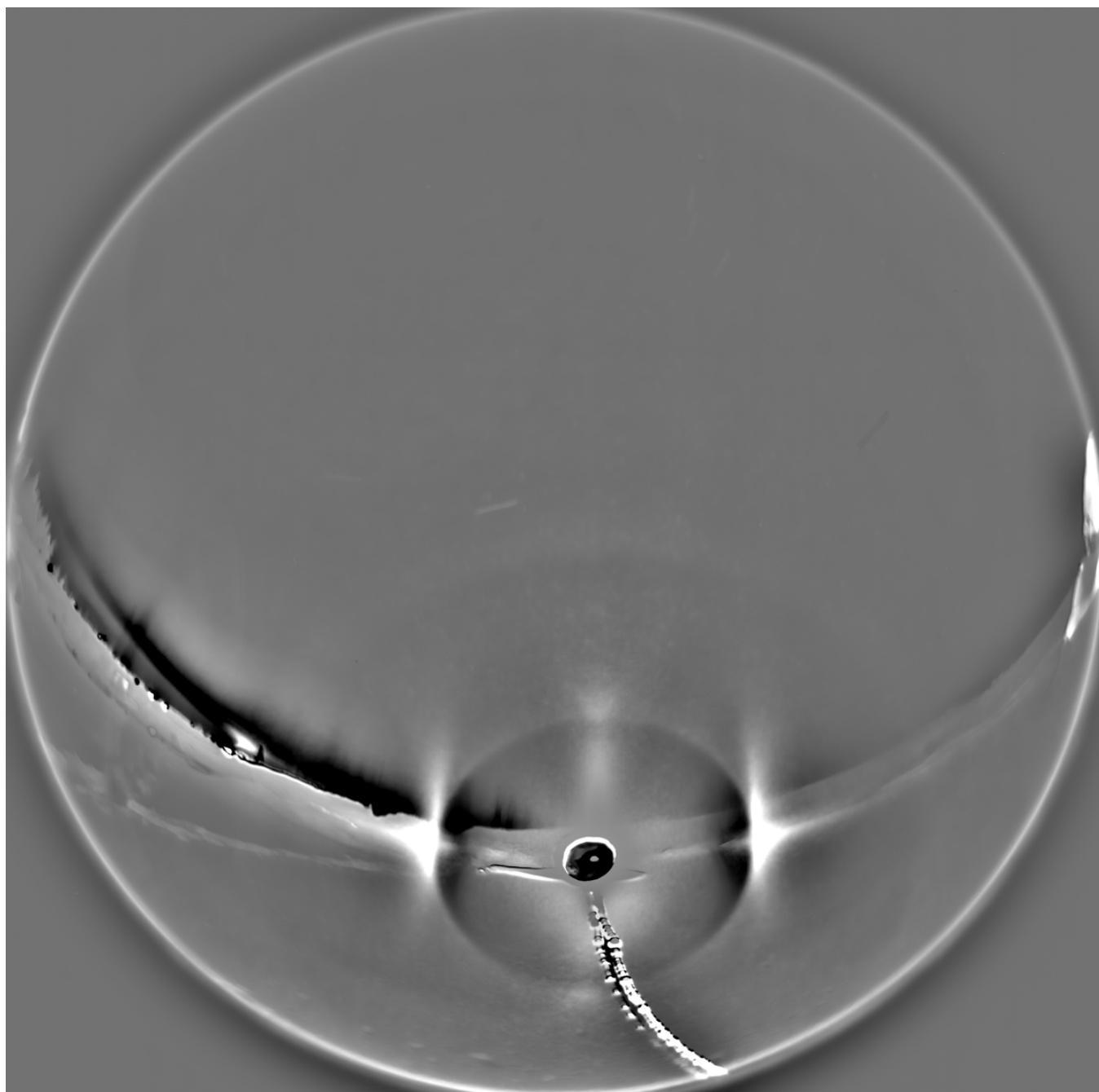
The thing is that there are a plenty of relatively thick segmented plates, potentially allowing for a raypath traversing adjacent branches through 60 degree prisms and thus producing a second parhelion at around 46 degrees. But there is no such animal in the photos. Most of the gapped crystals are rather irregular and were probably not that stably oriented.

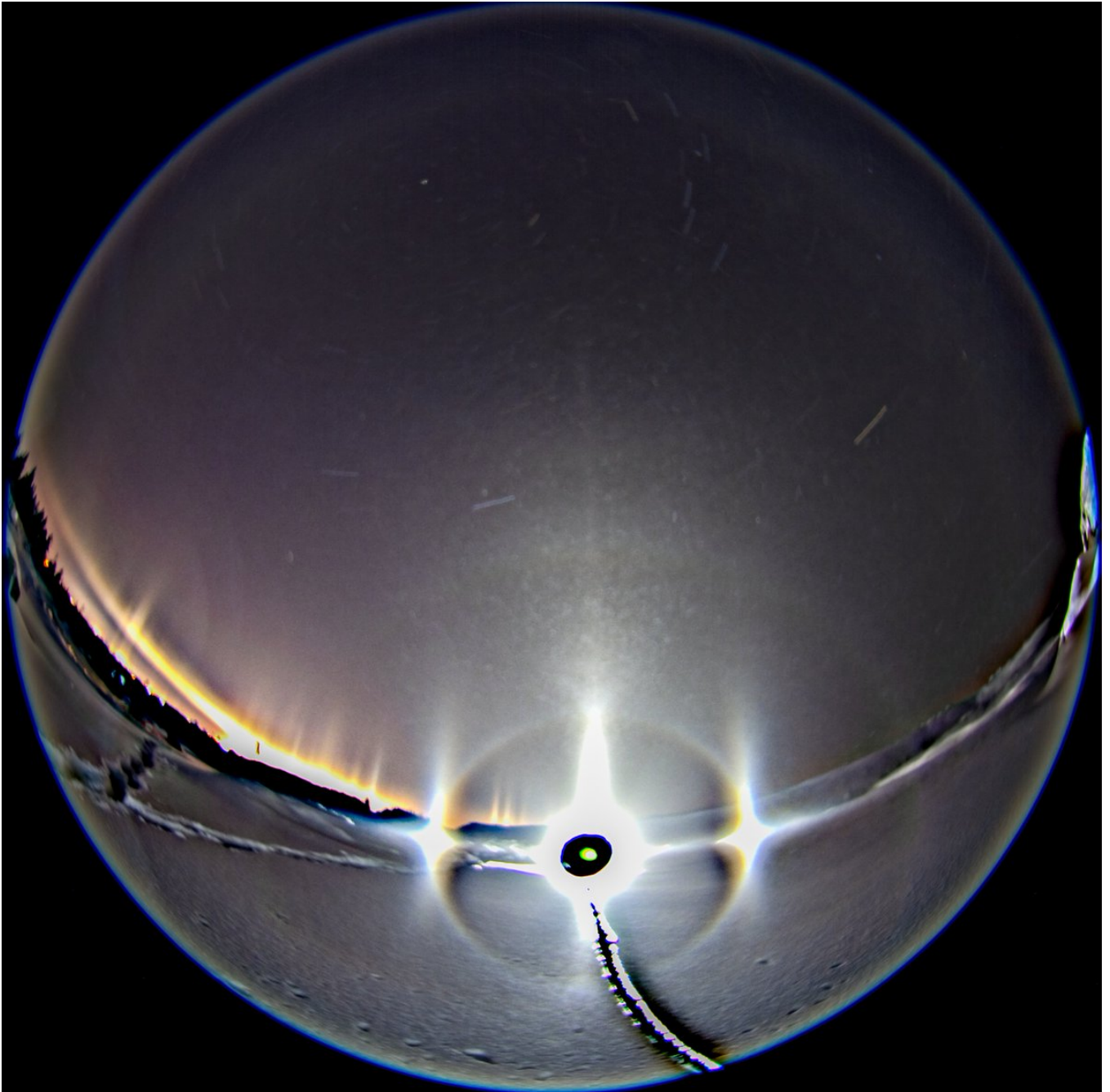
Second interest is in the tangent /Parry arc. There is some tiny columns in the samples, some in focus, others out of focus (I was not thinking of them when taking photos) but much more probably would have been expected. So we are talking about a sampling bias here: very small crystals have tendency just to hover about and don't land so easily on the dish as the bigger crystals.

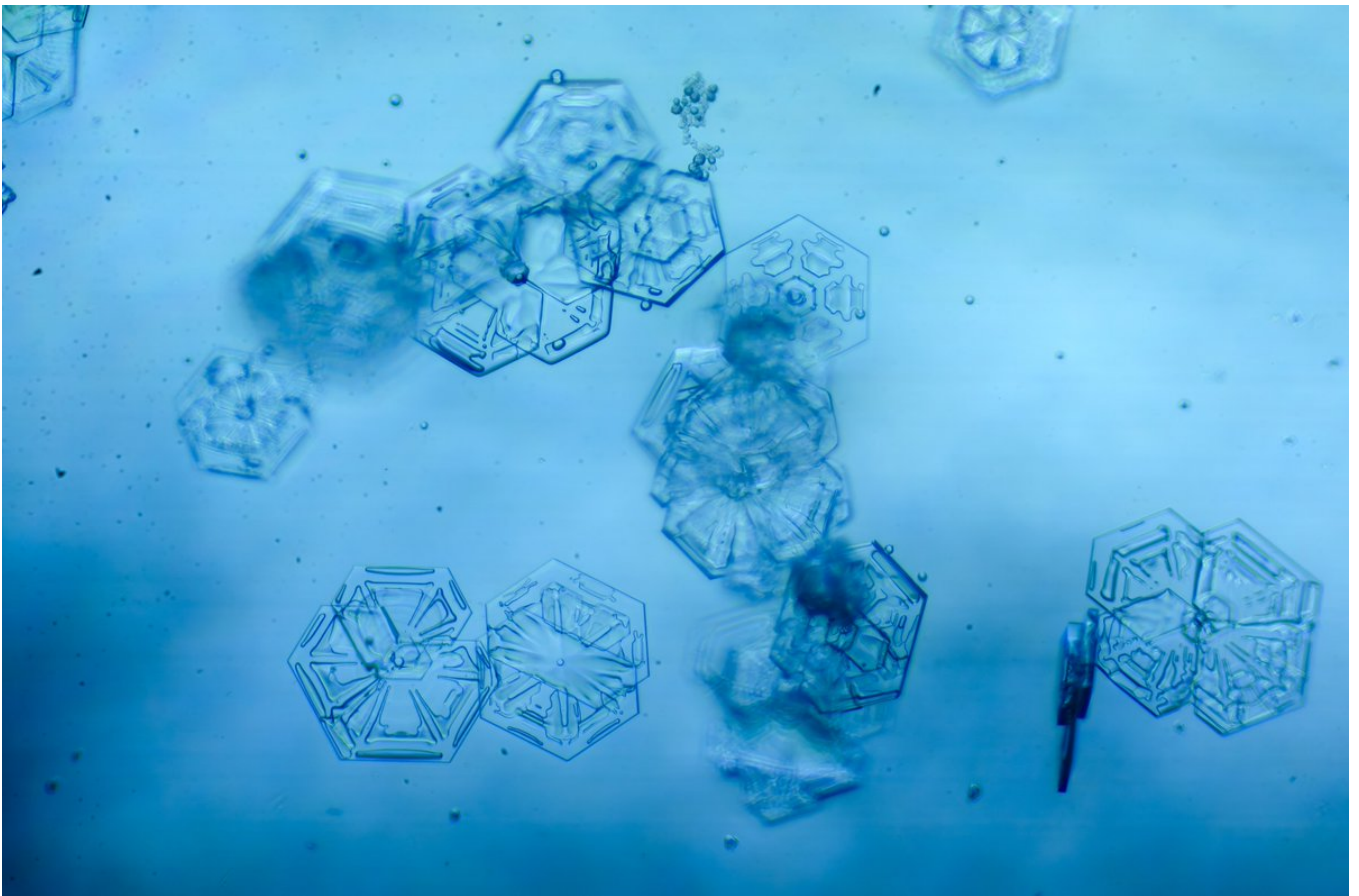
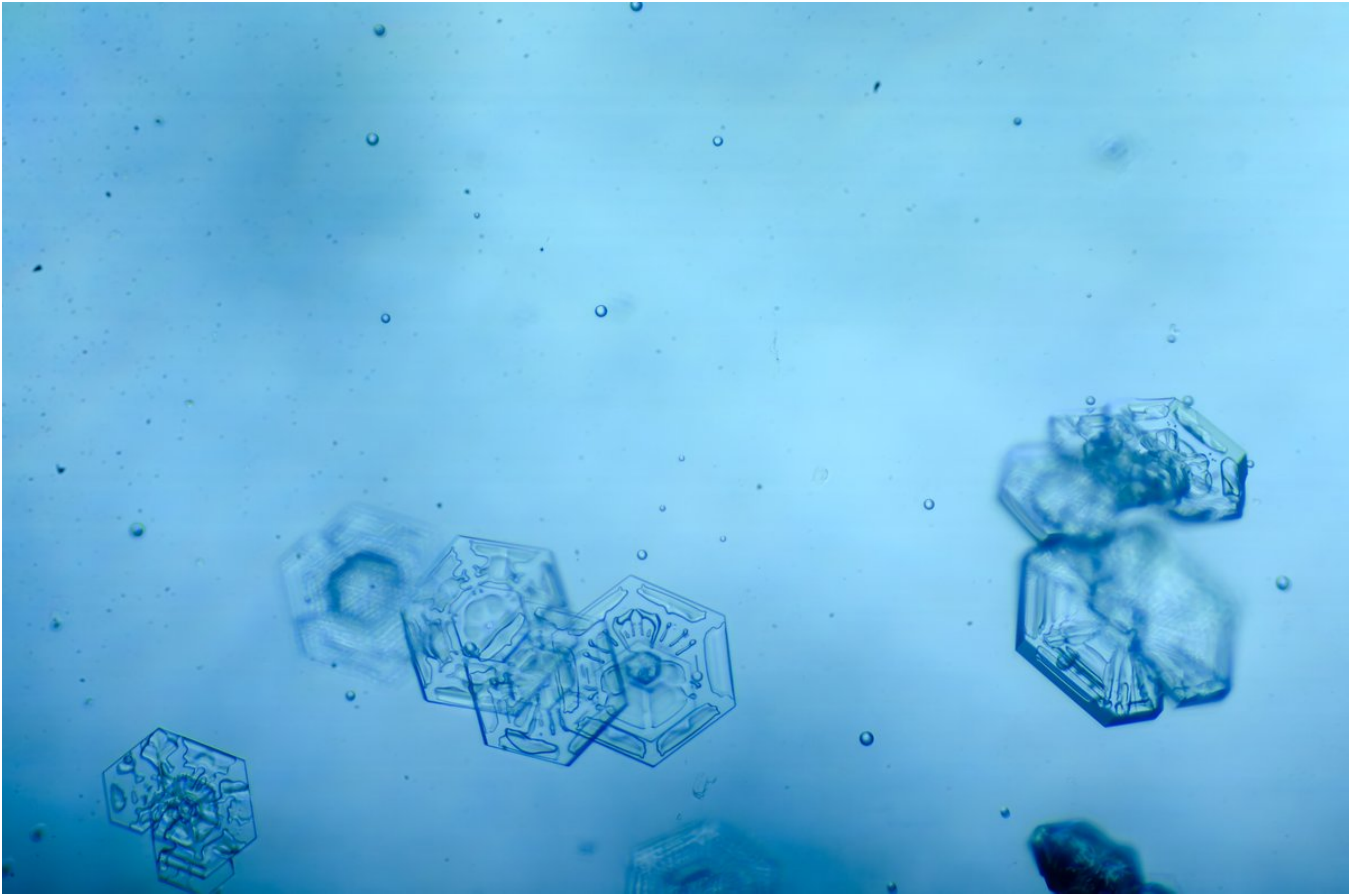
But it is also possible there were more those crystals in the dish but majority of them were on such level as to make them completely out of focus and thus practically hidden from the inspector. This is a real feature that I should always remember. The charcoal lighting fluid I use results often in three layers: near the surface, bottom and somewhere in between. These should be always checked.

So that's another source of sampling bias. In what proportion these two biases played part here I venture not to say. I will return to this issue at some later point as I have a good case study from another display.

By the morning the temperature had settled at -32 C in my thermometer.

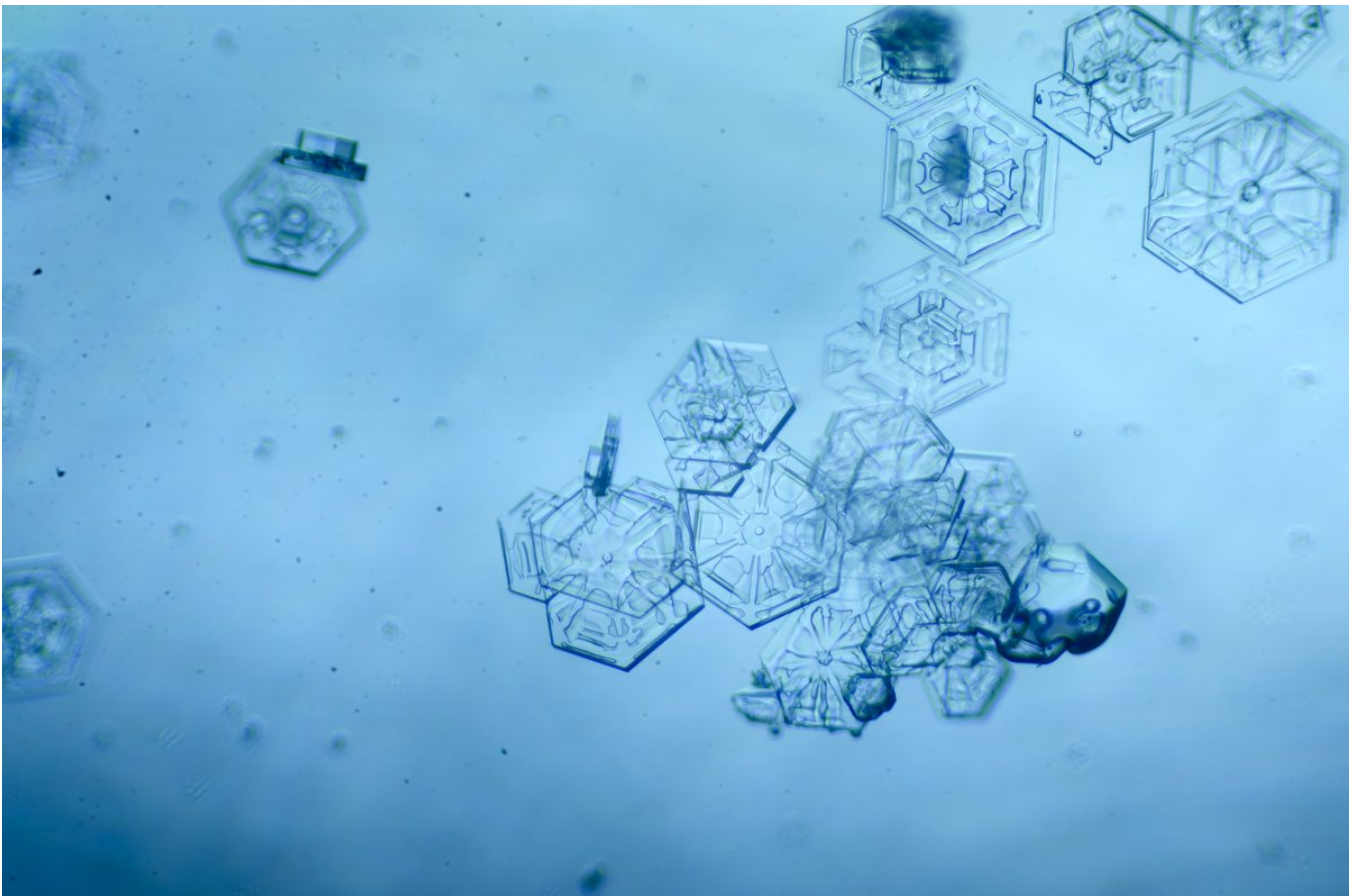
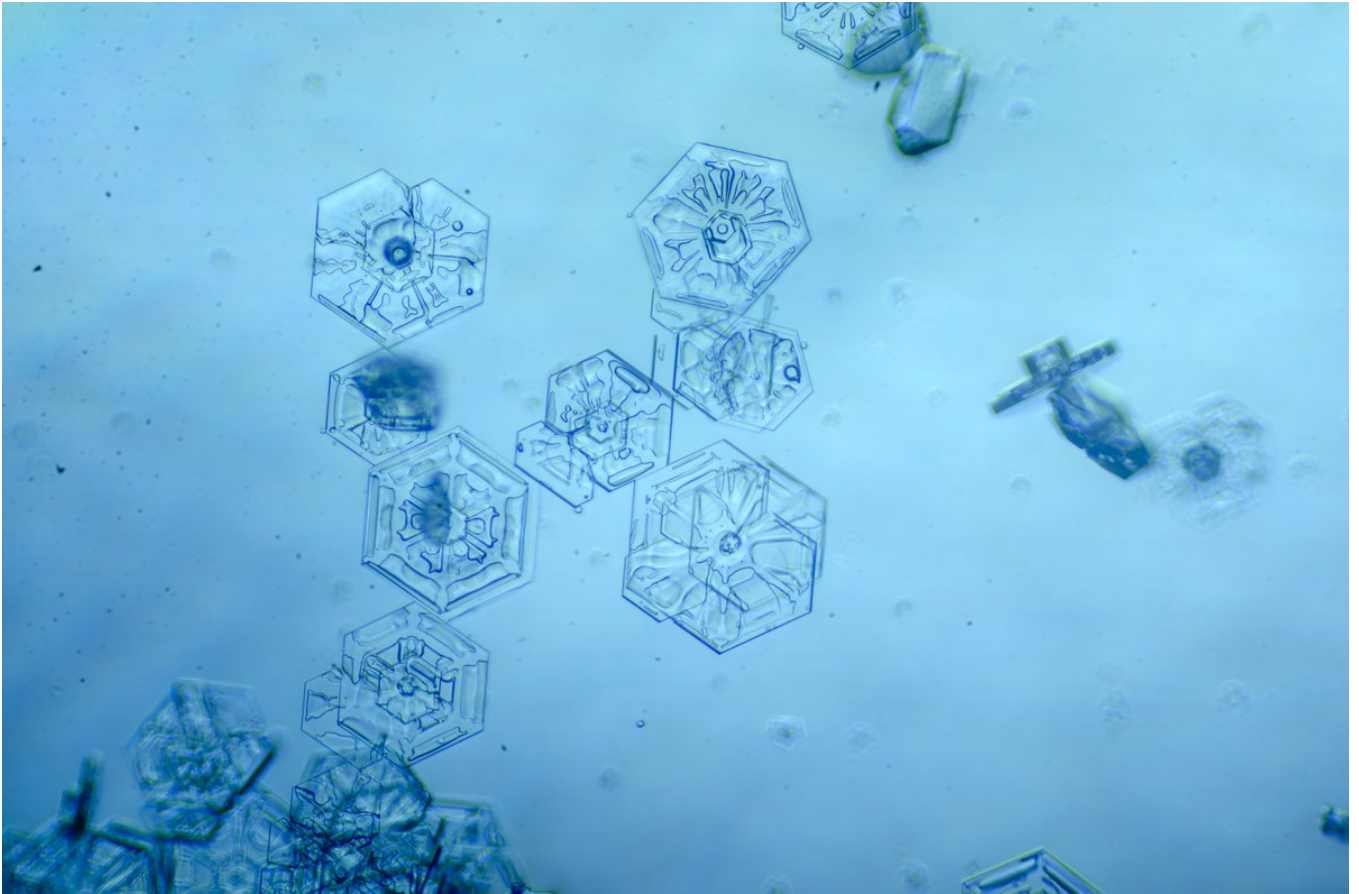


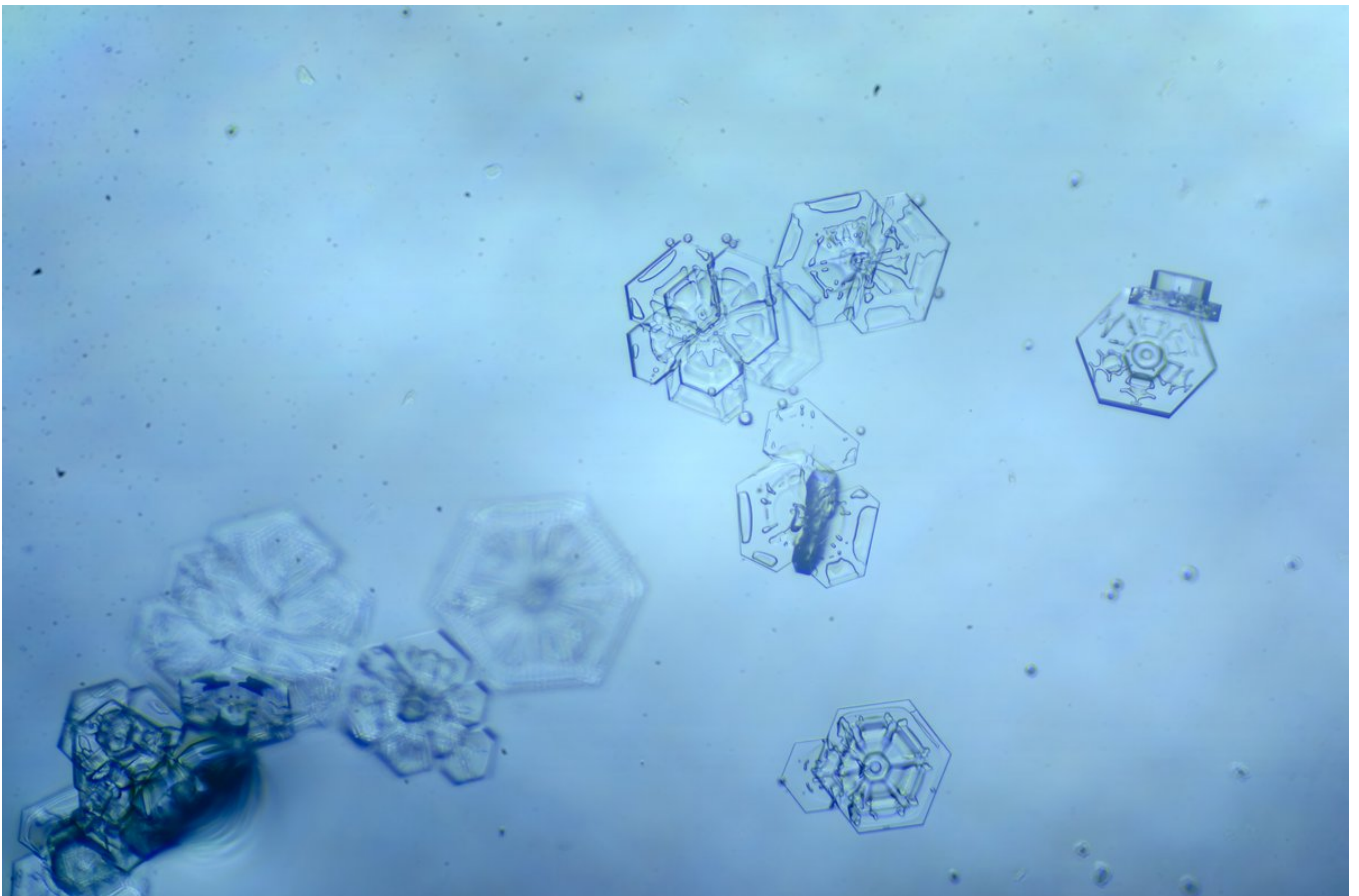
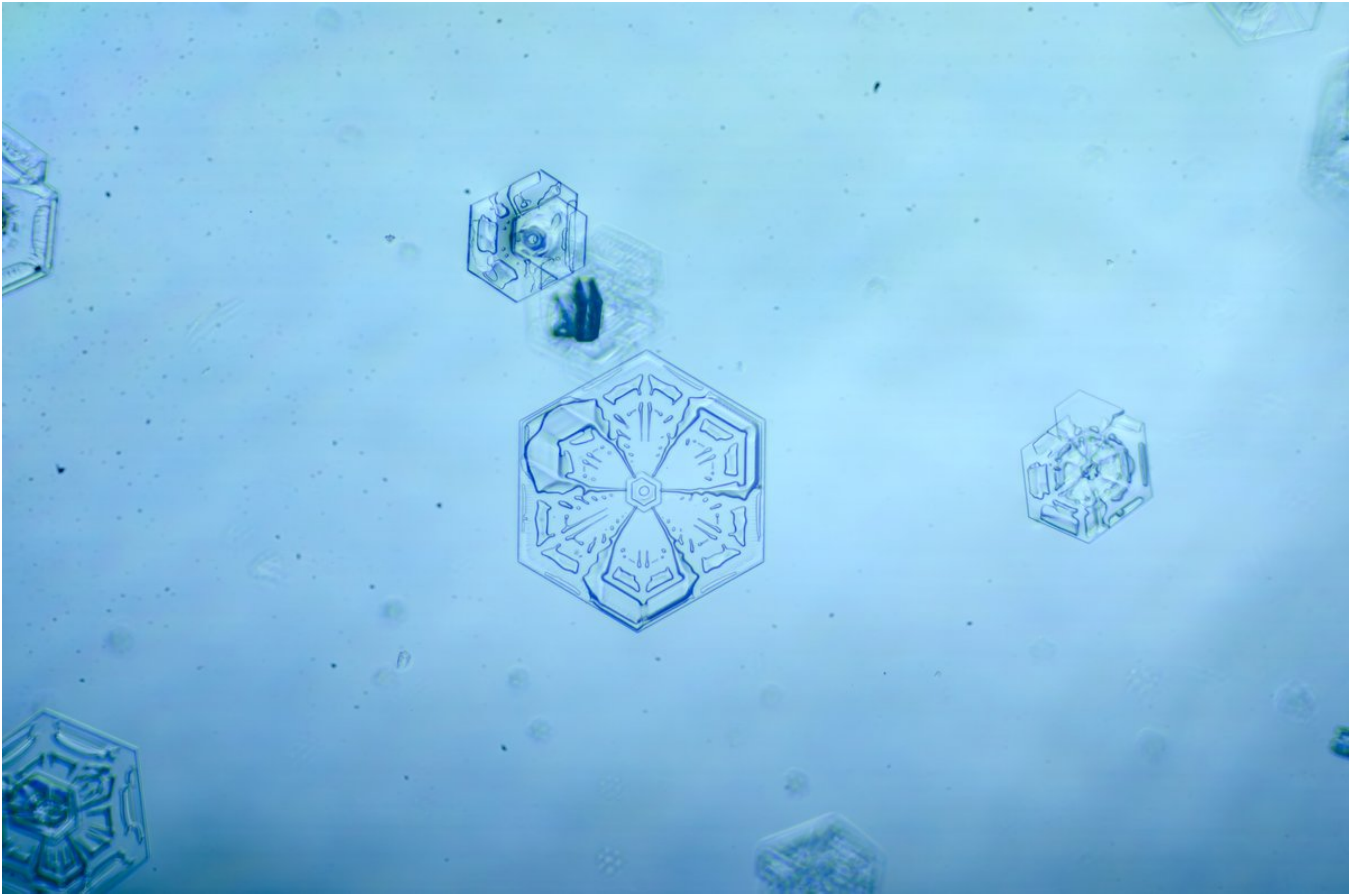




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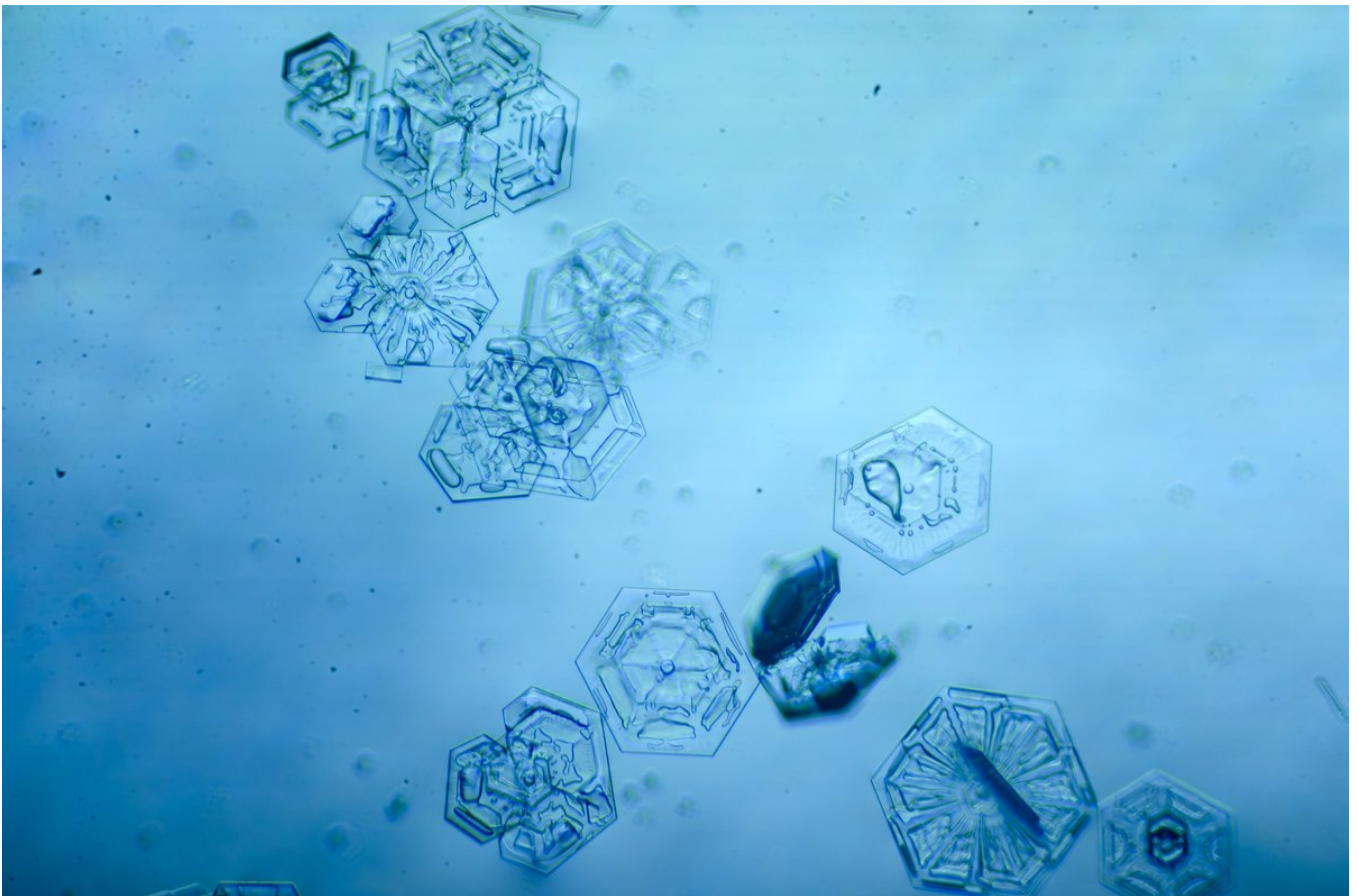
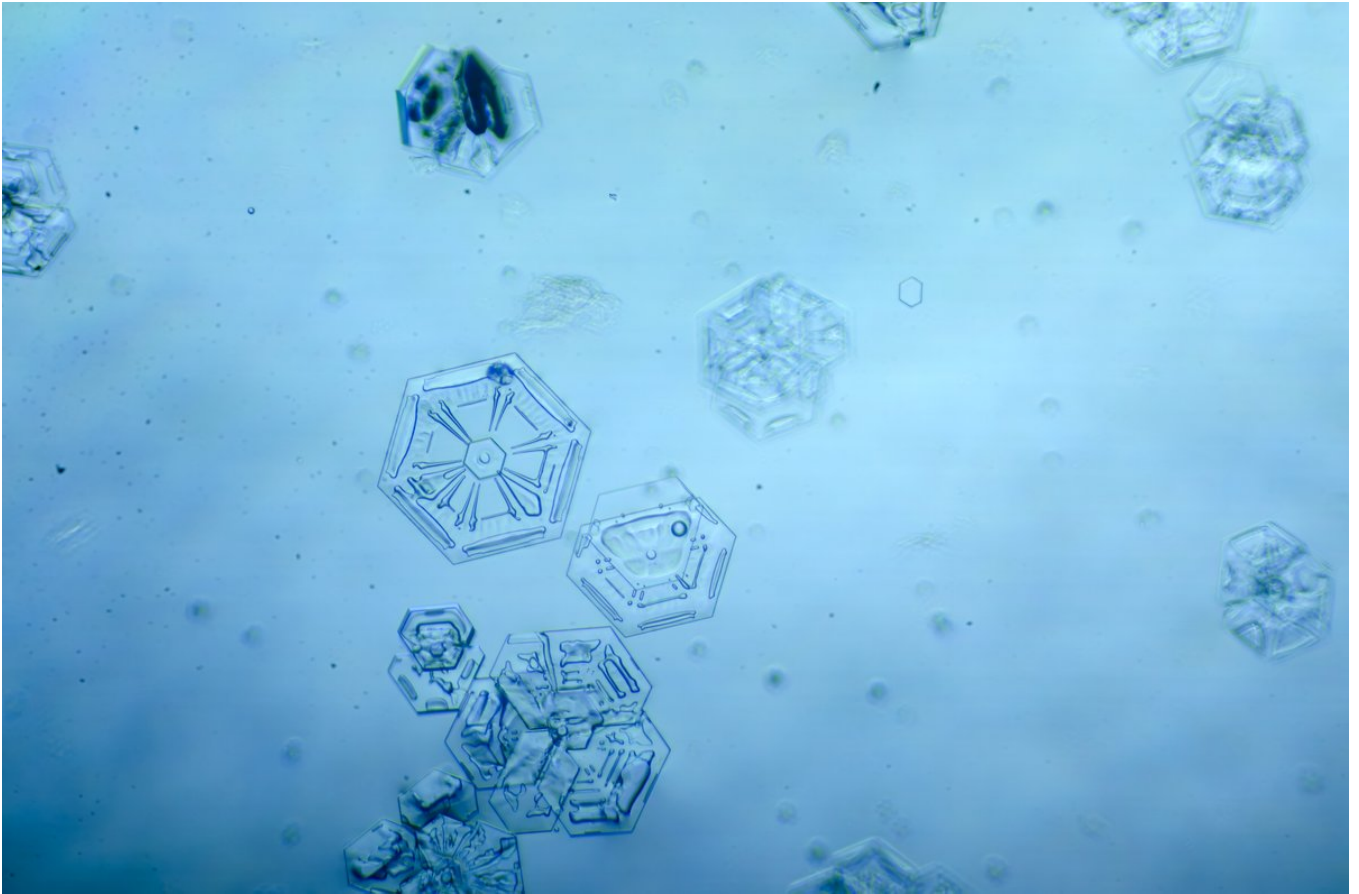
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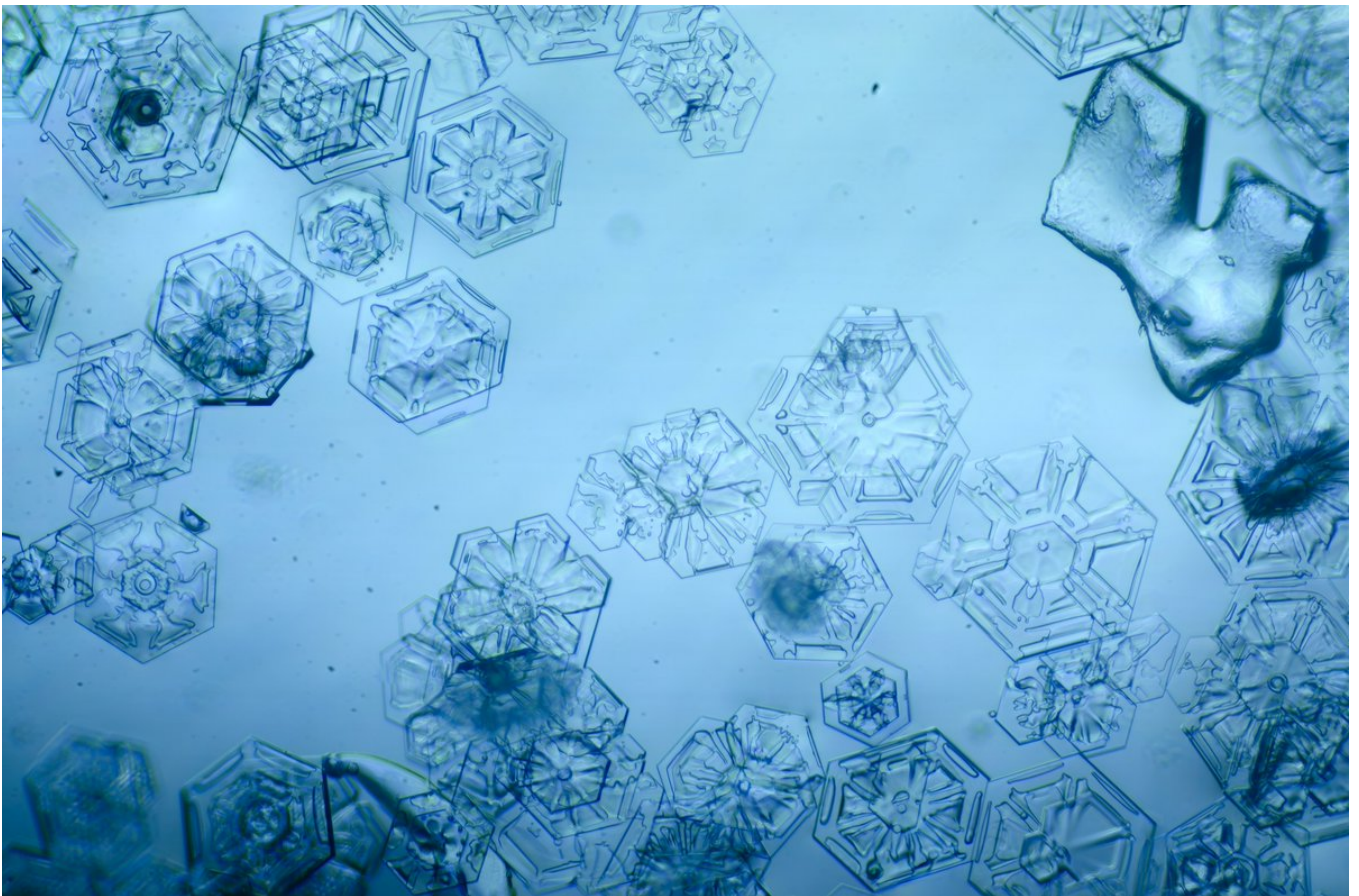
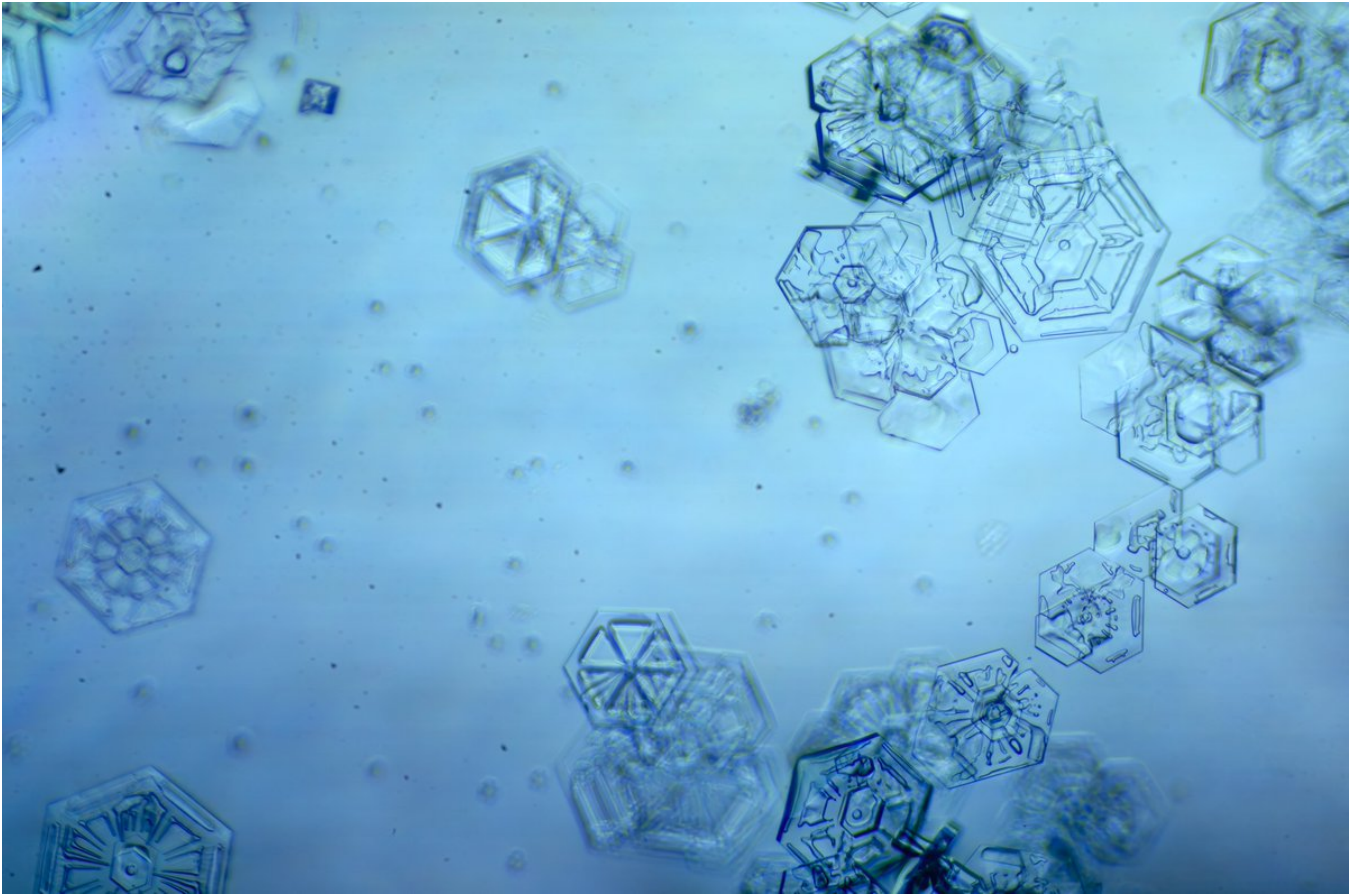




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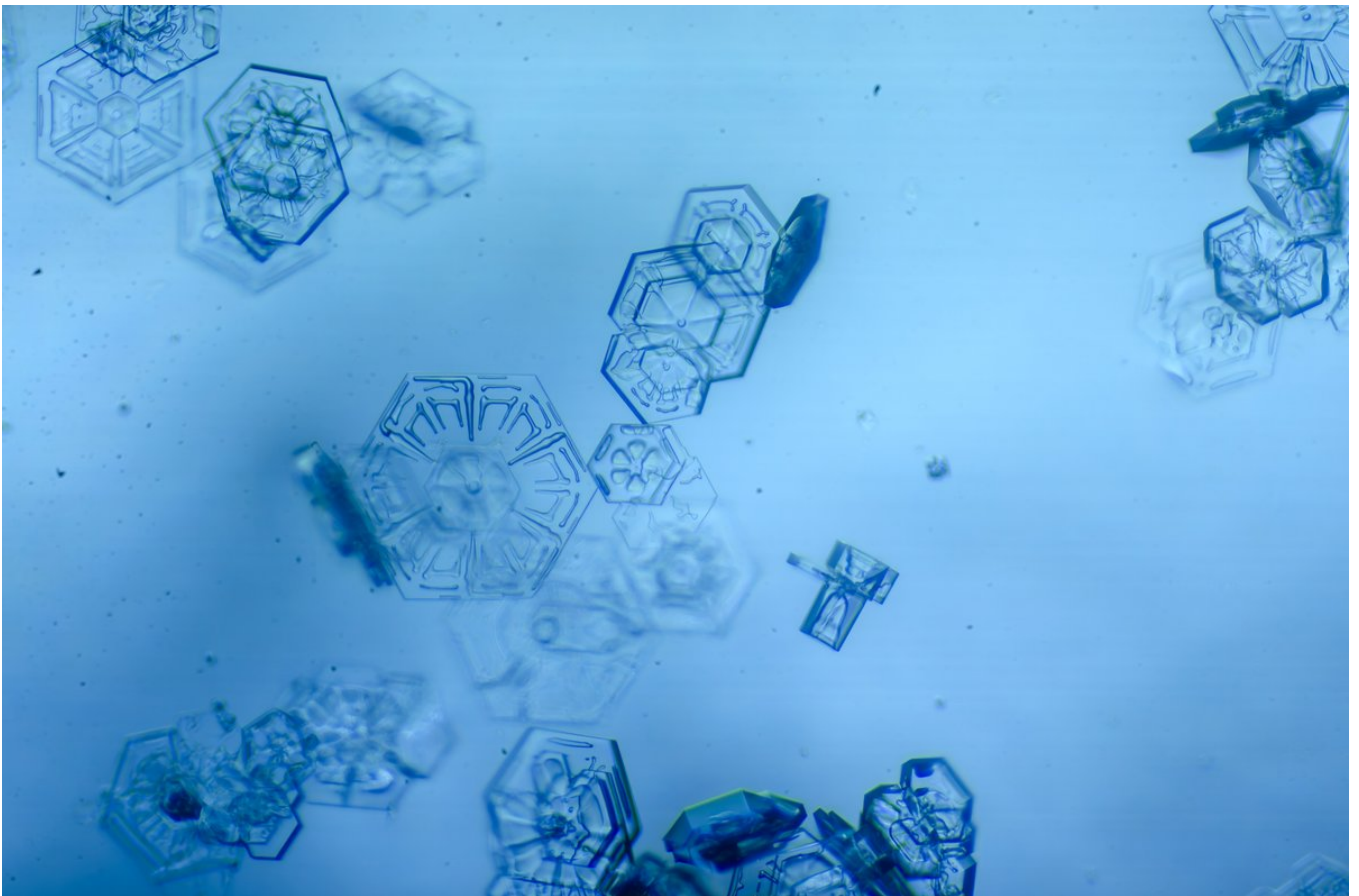
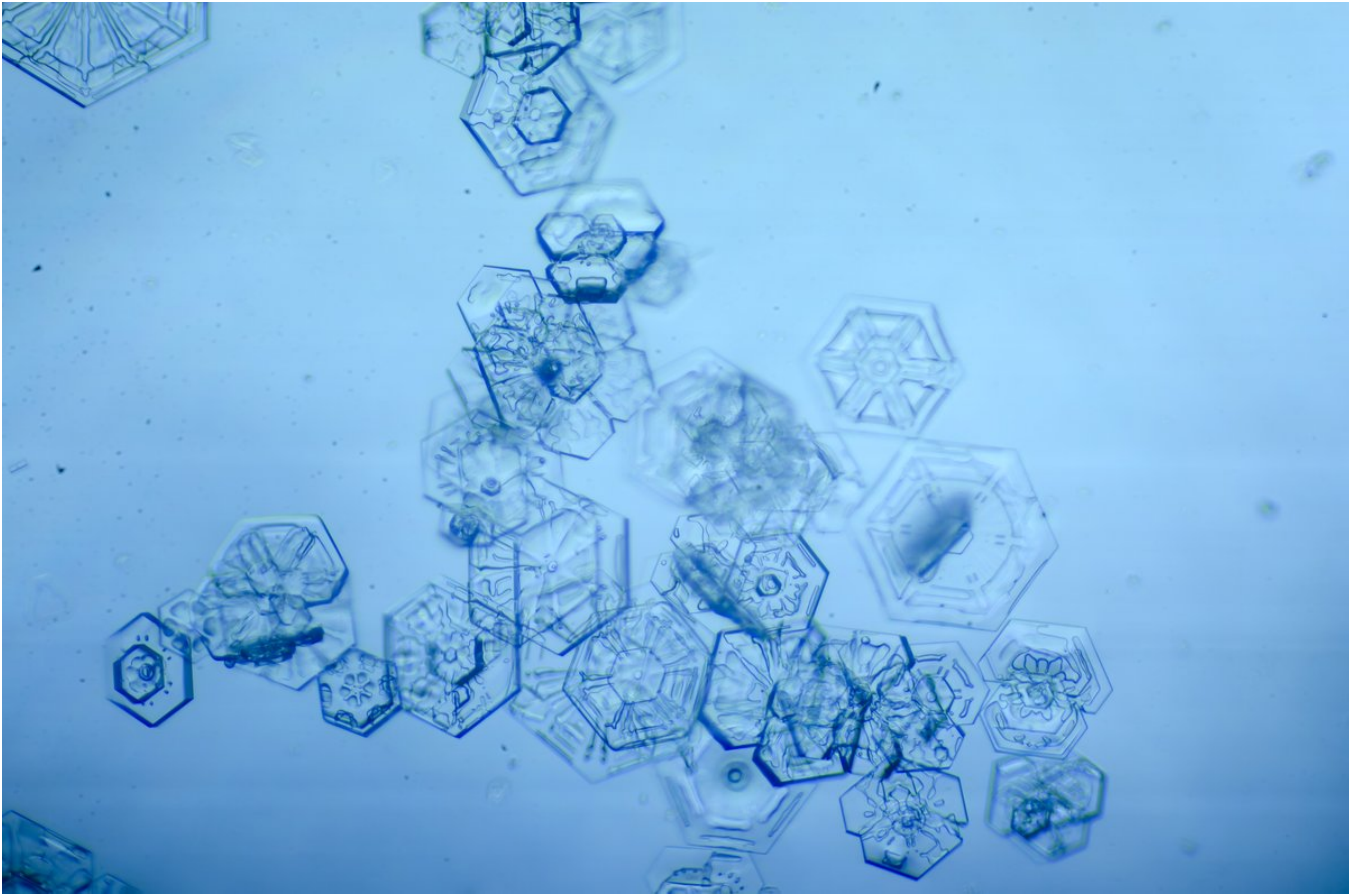
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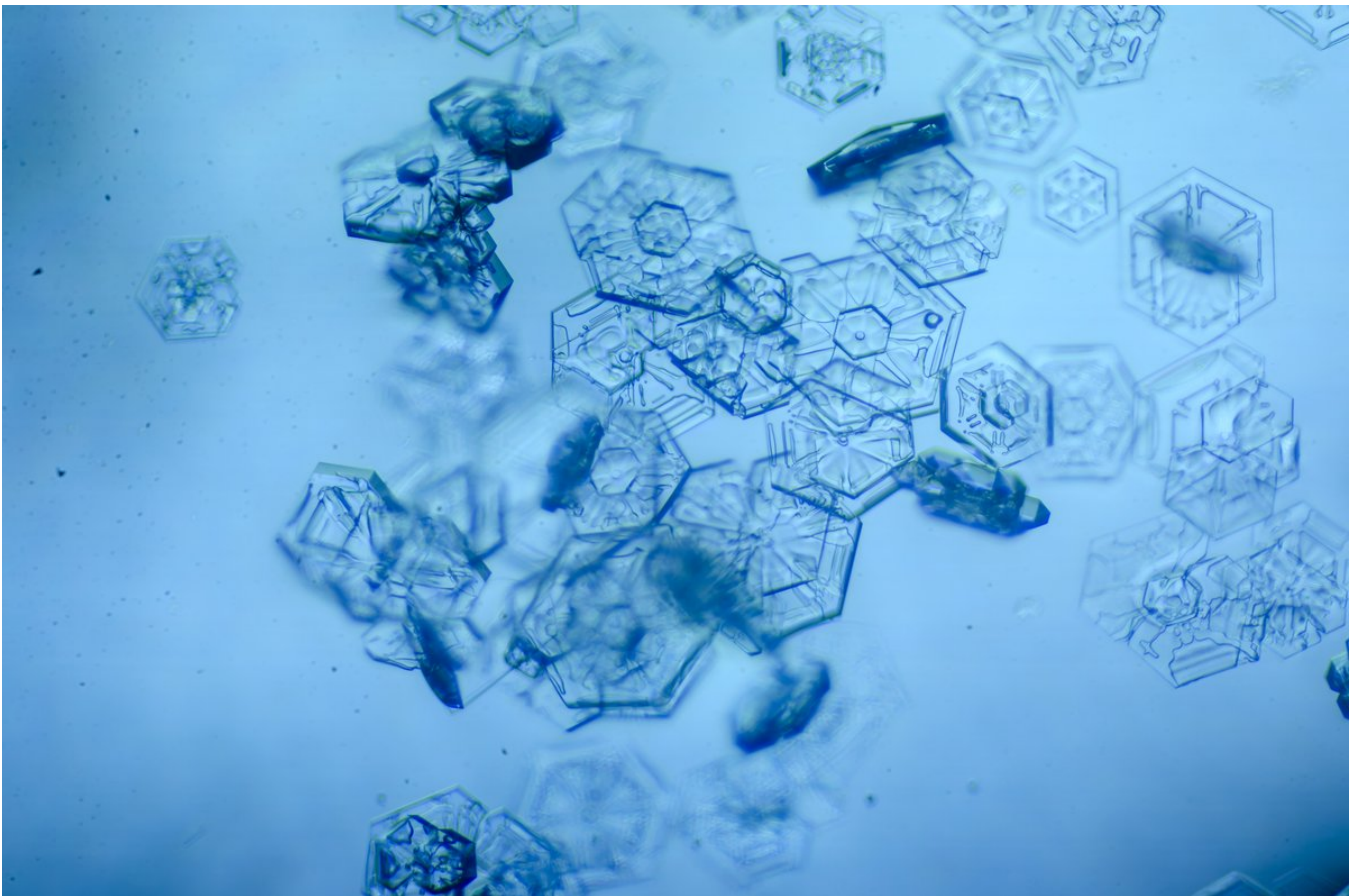
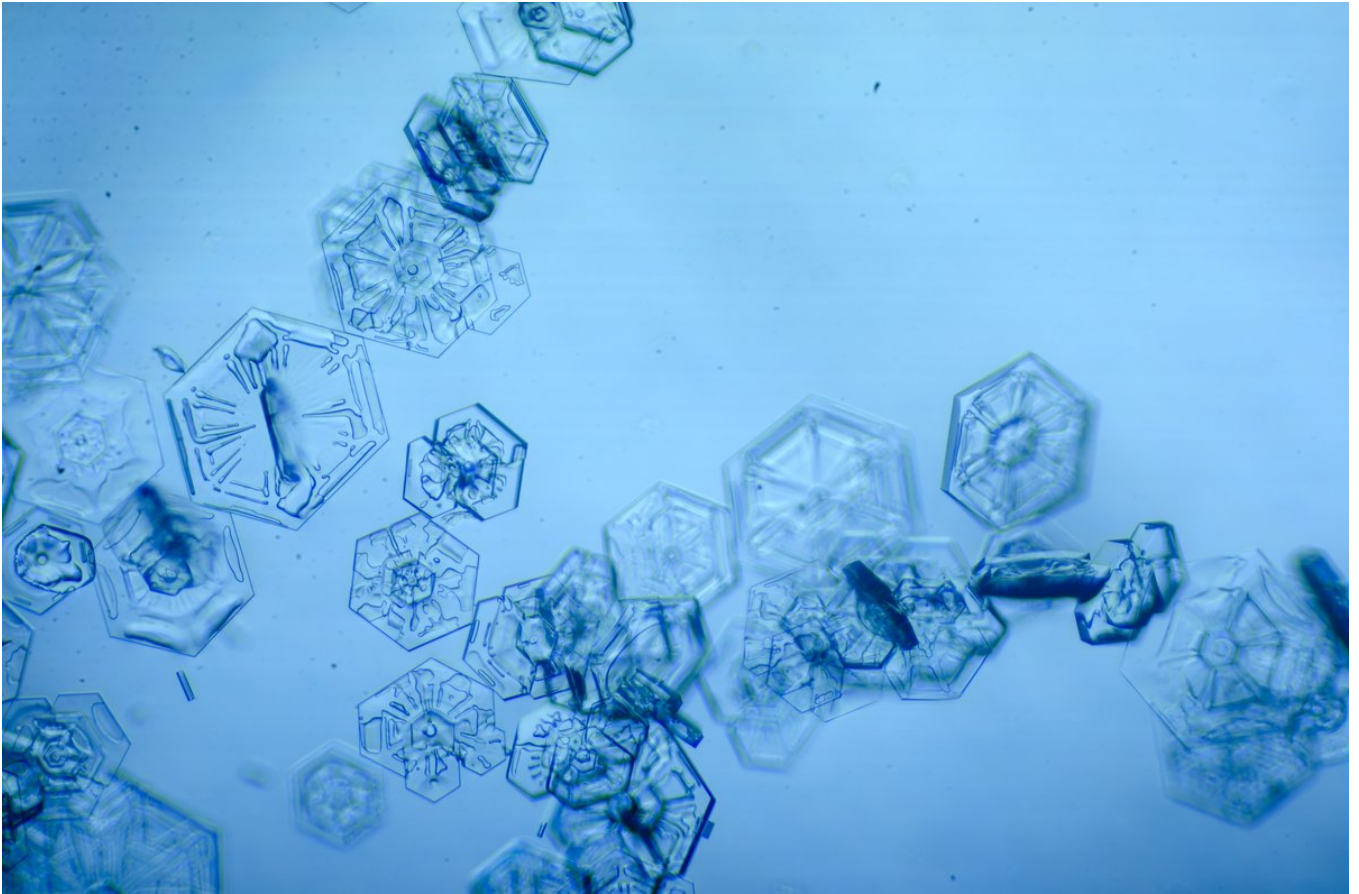




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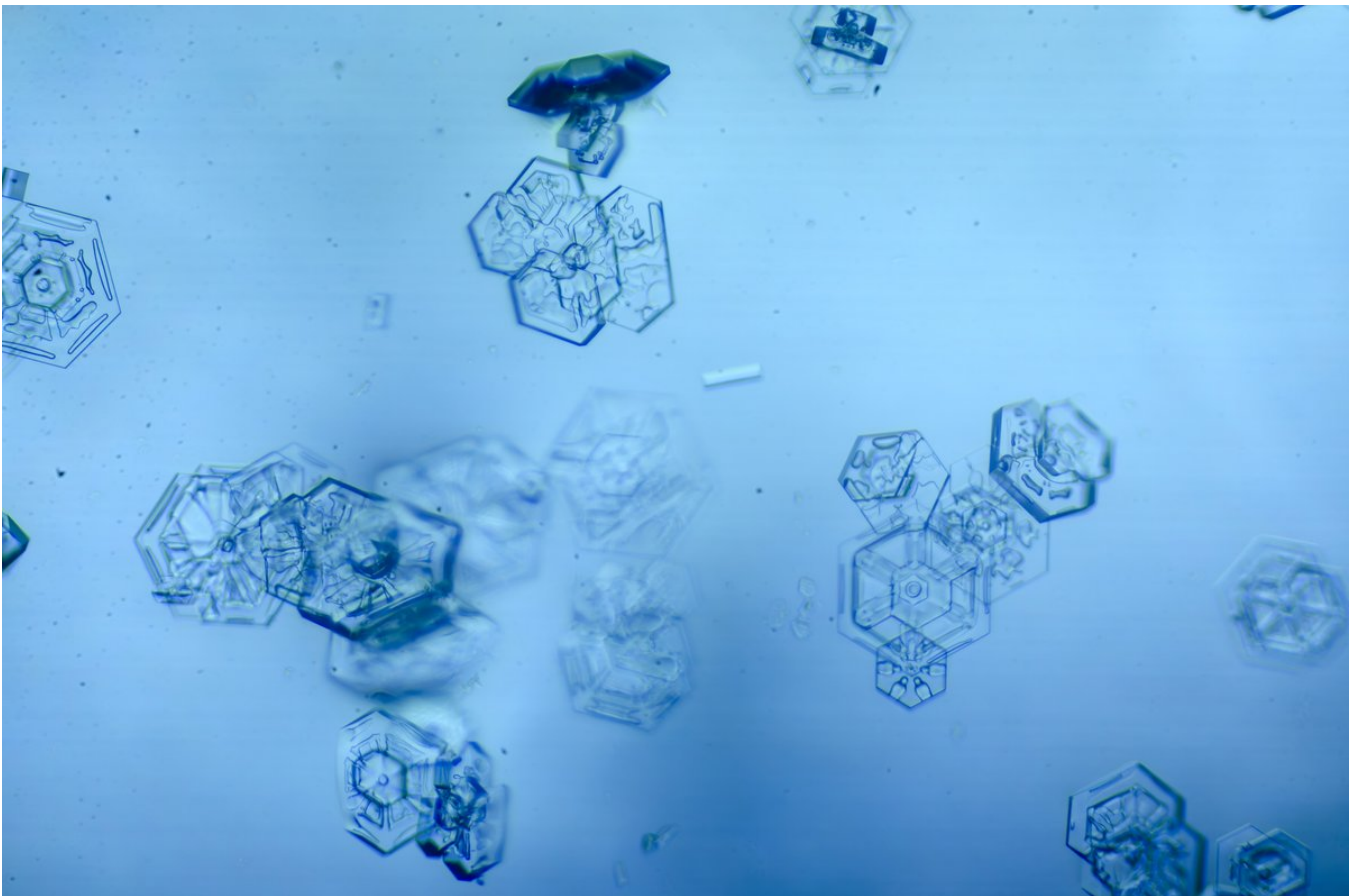
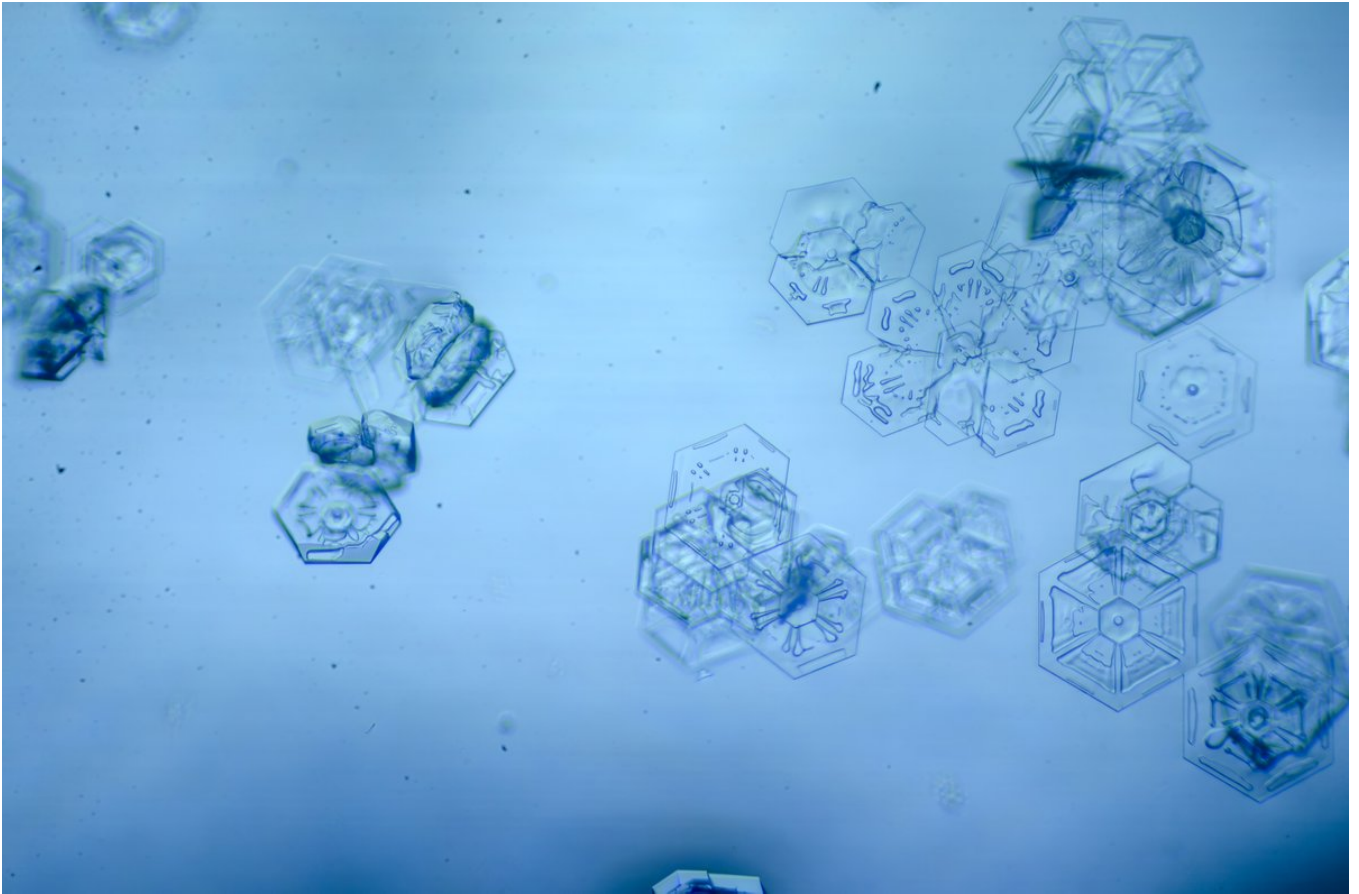
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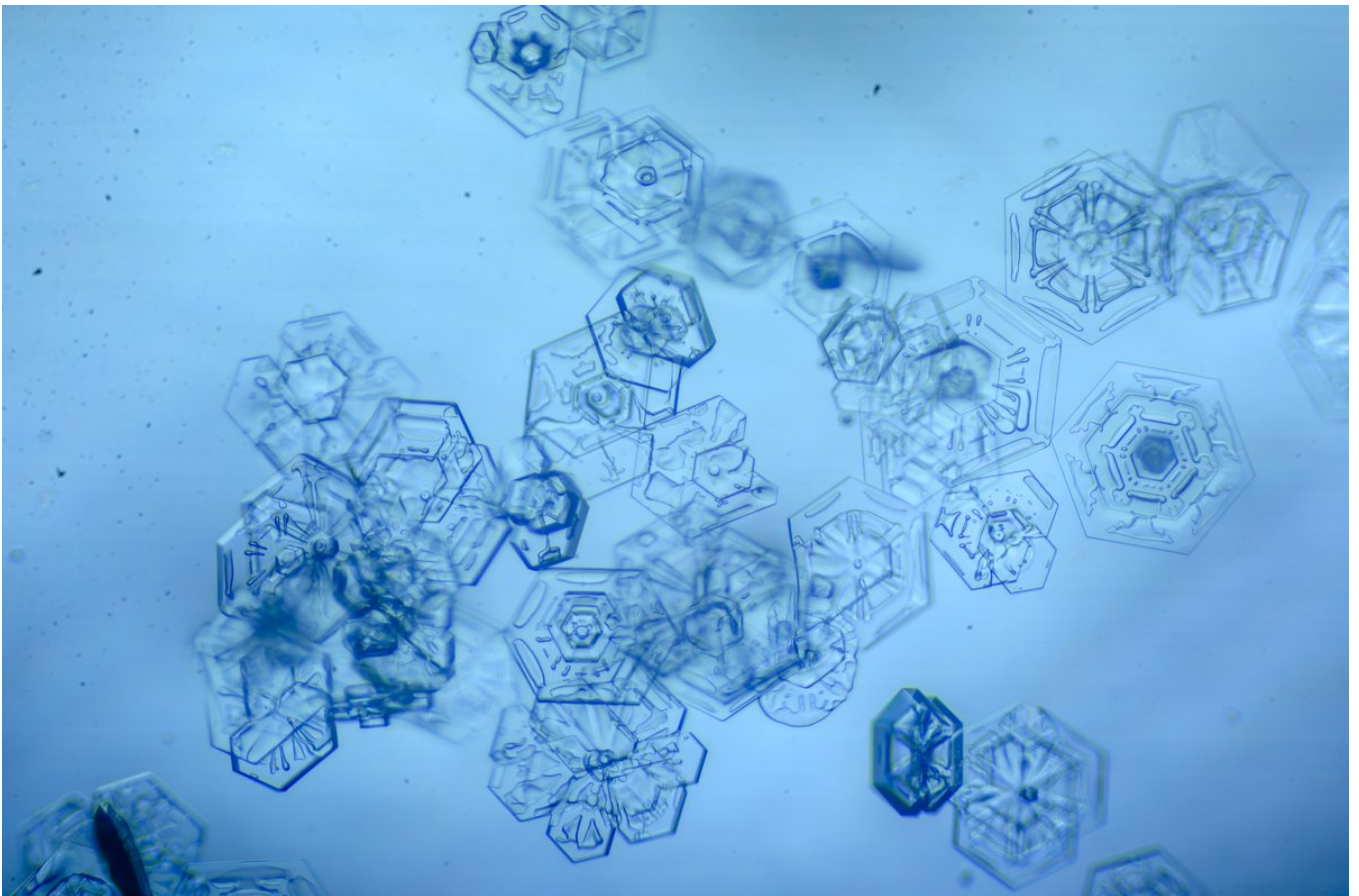
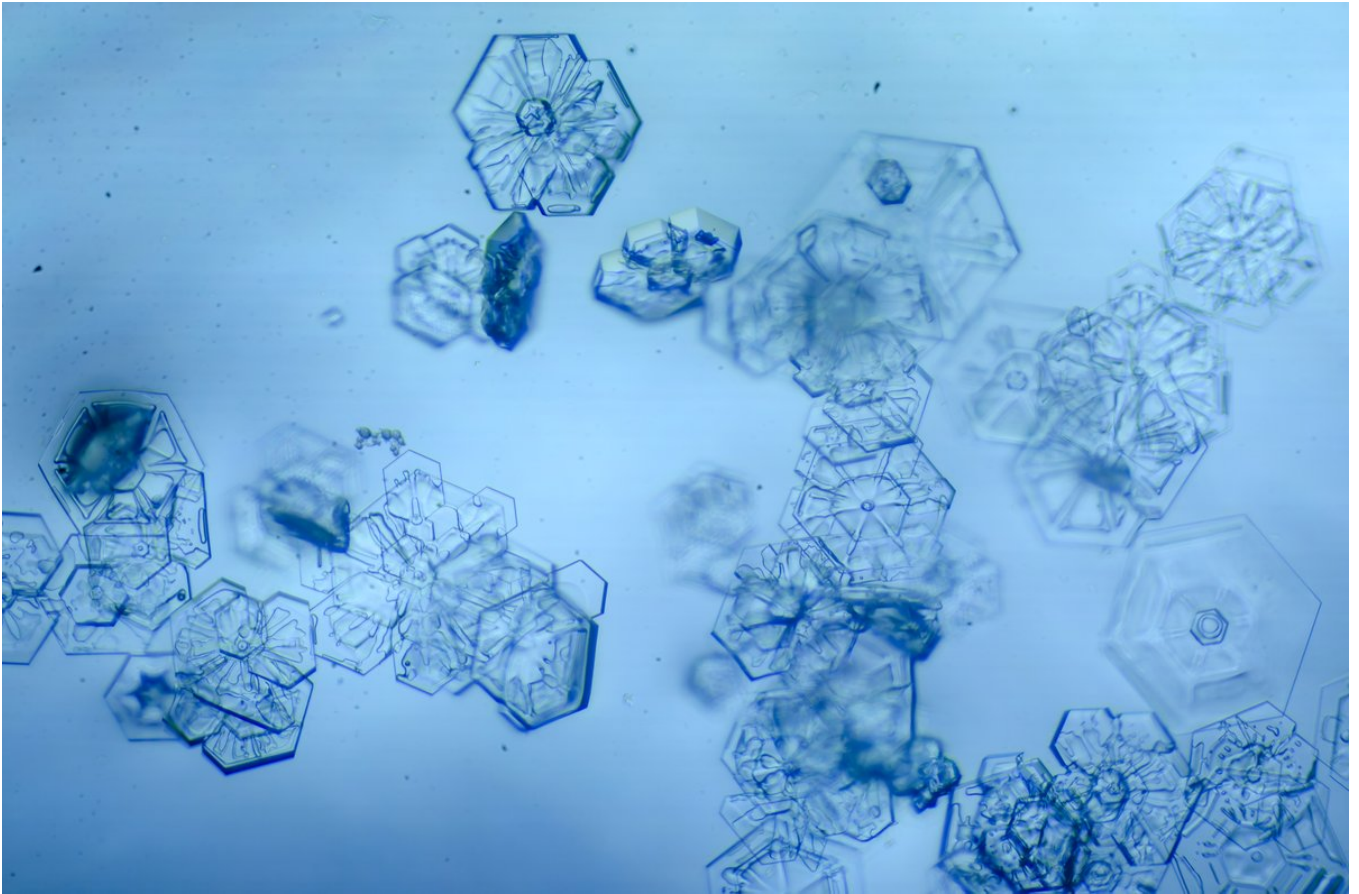




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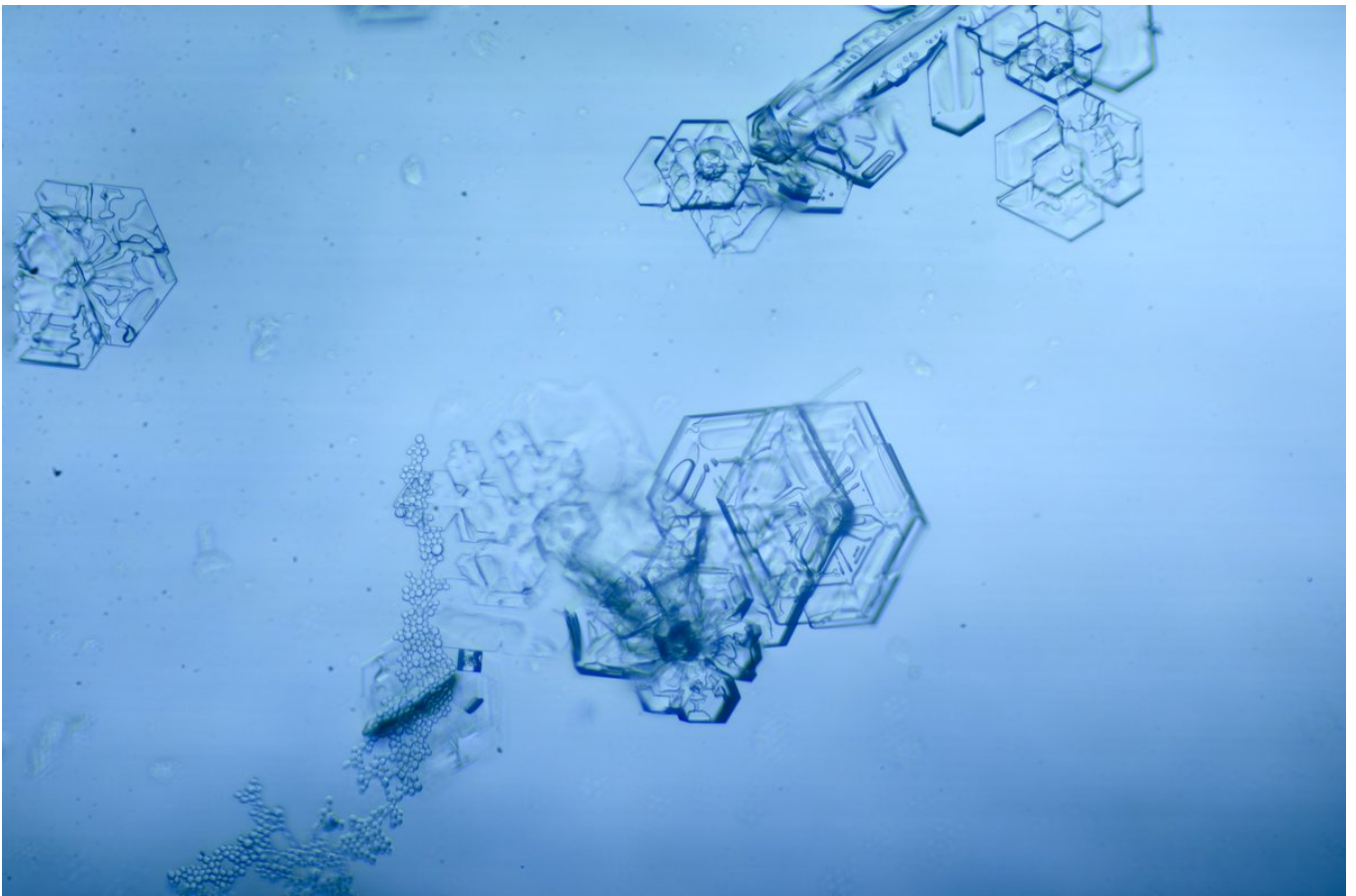
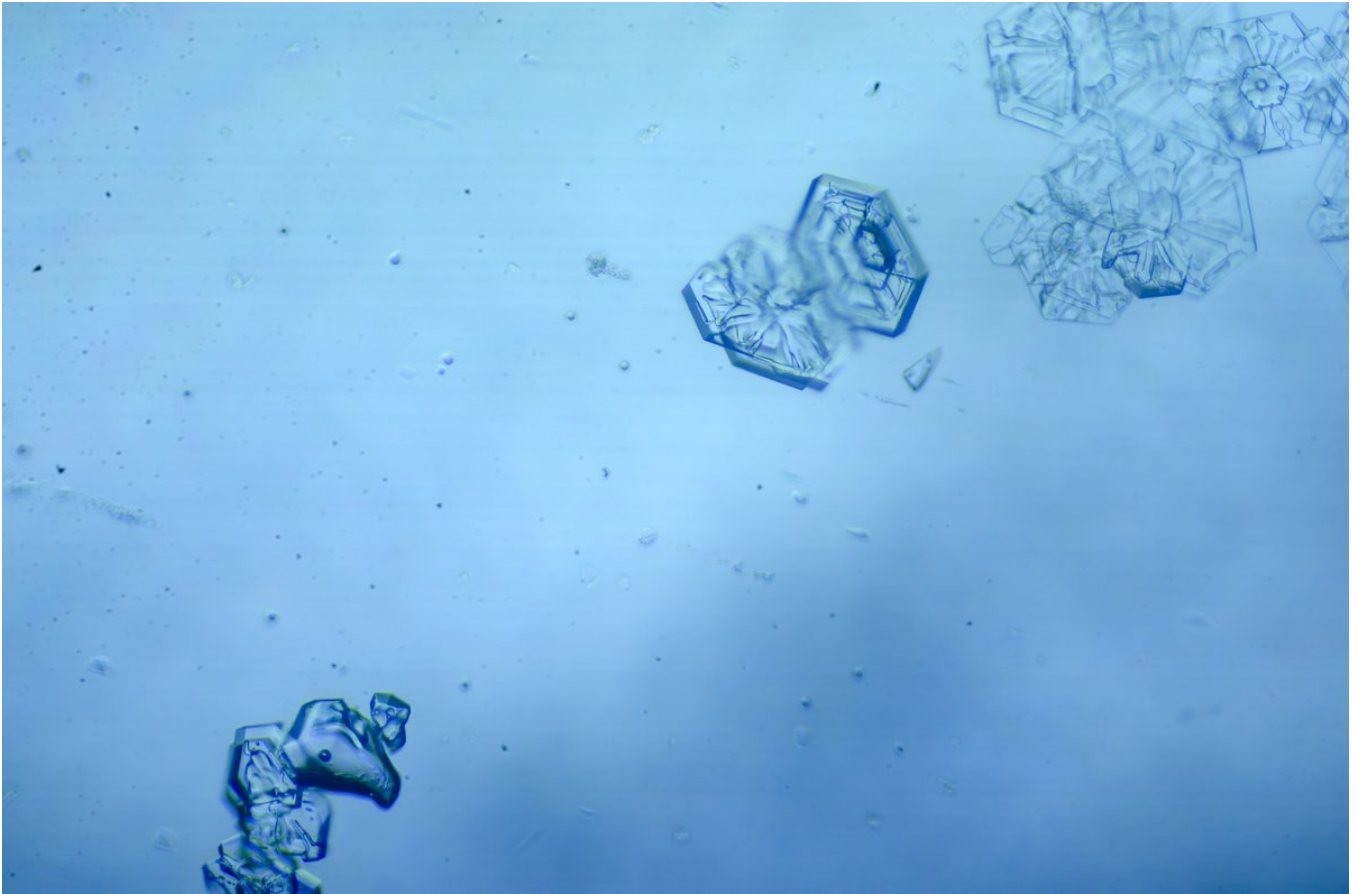
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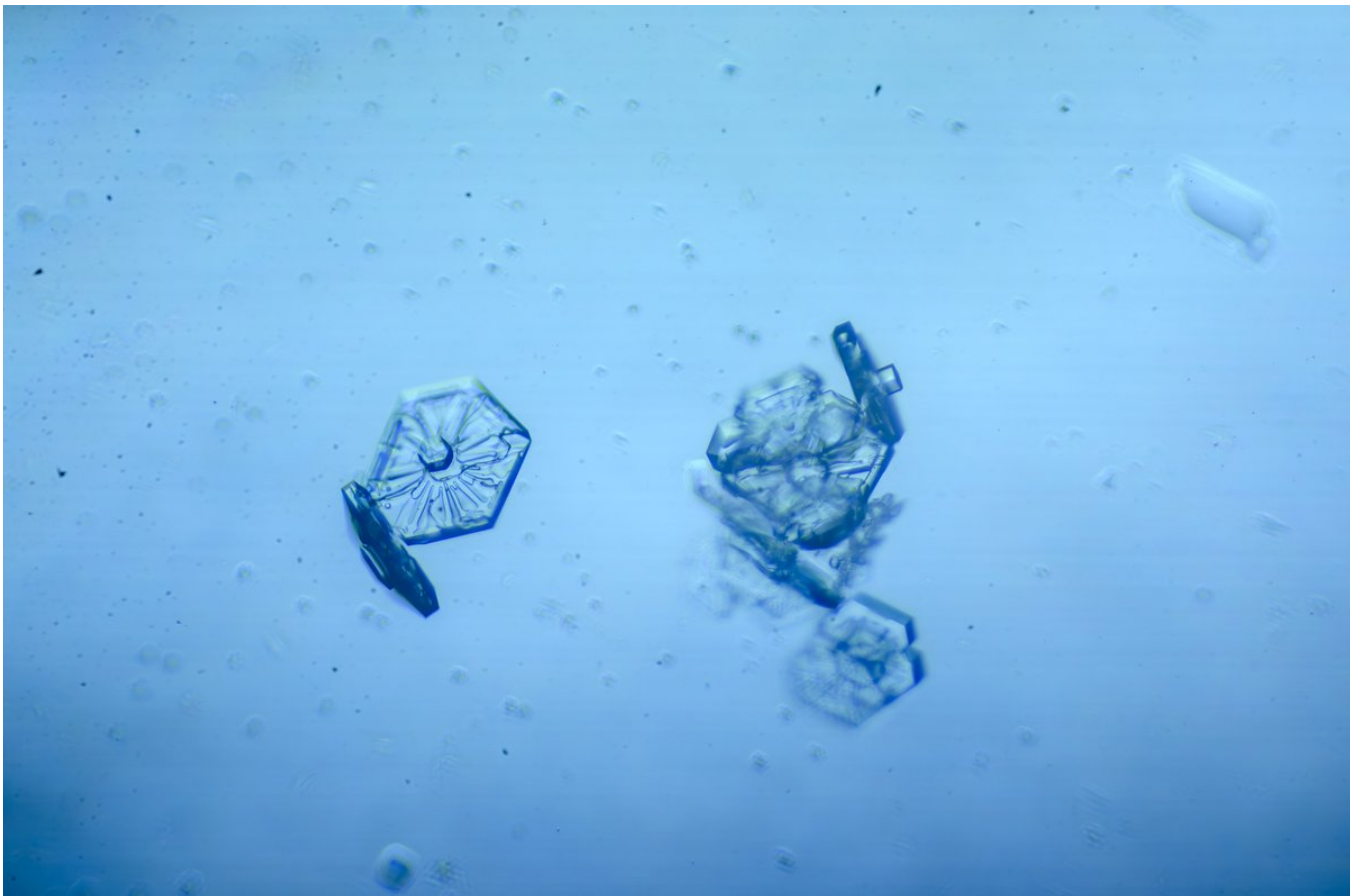
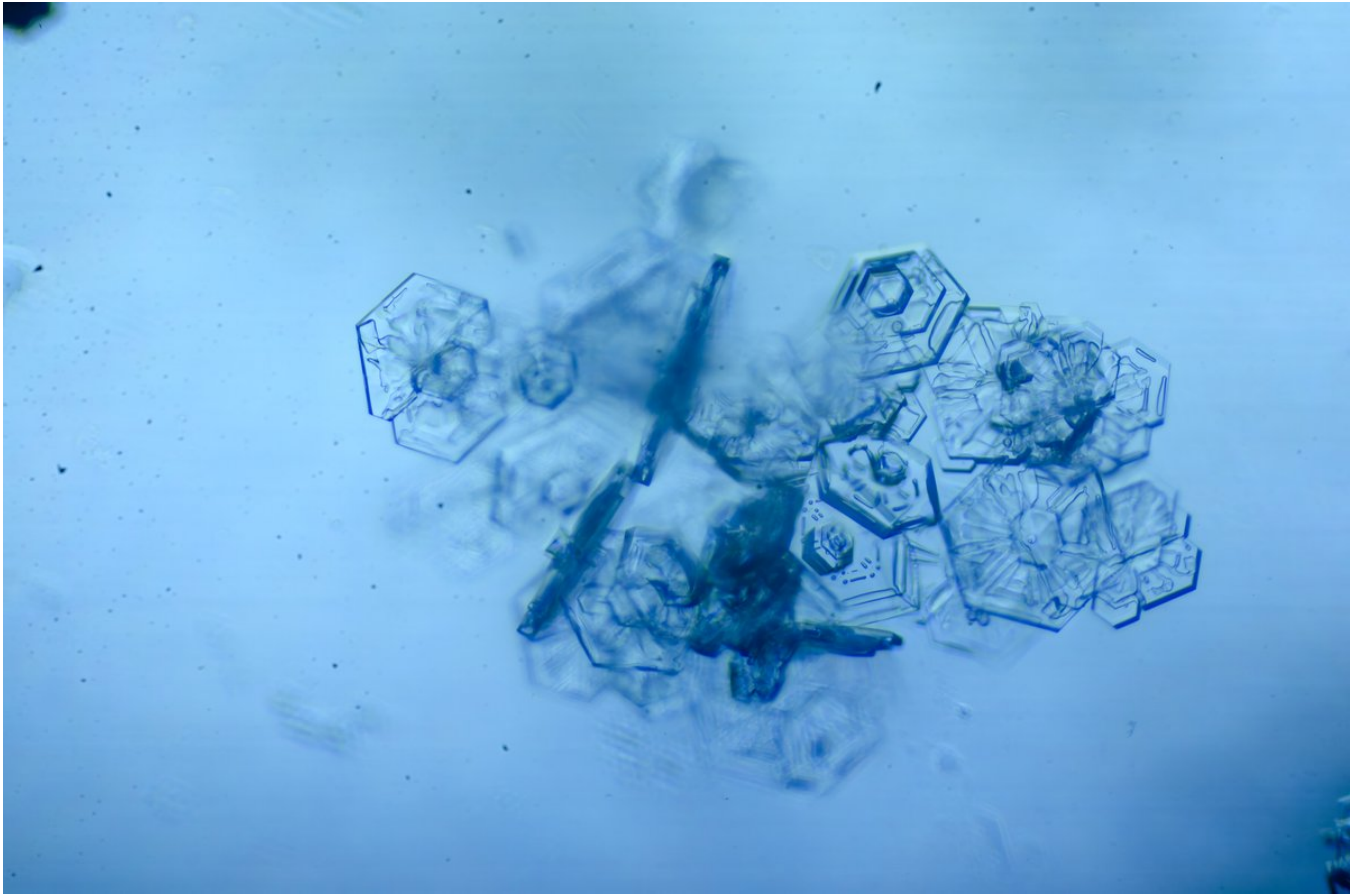




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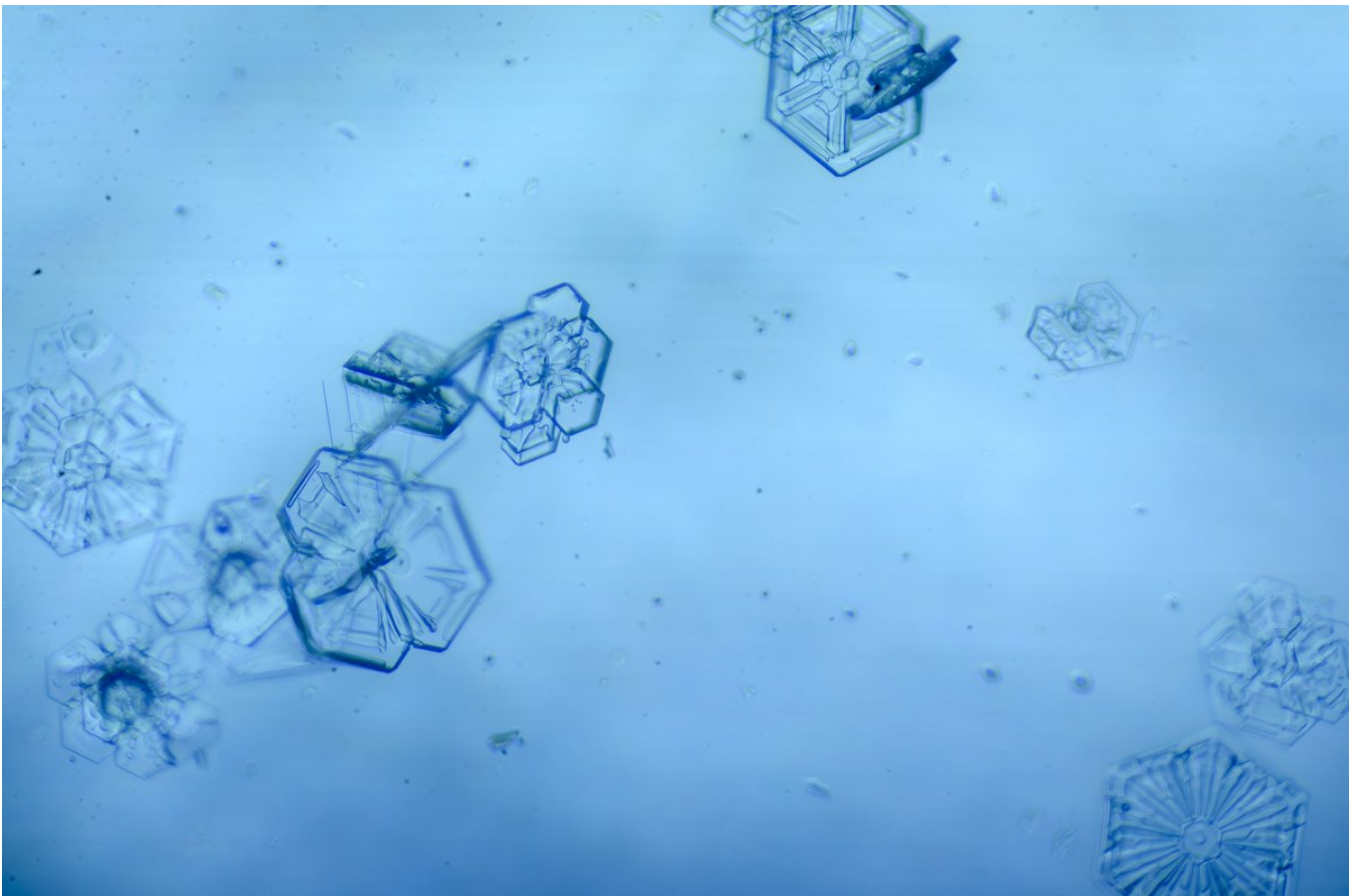
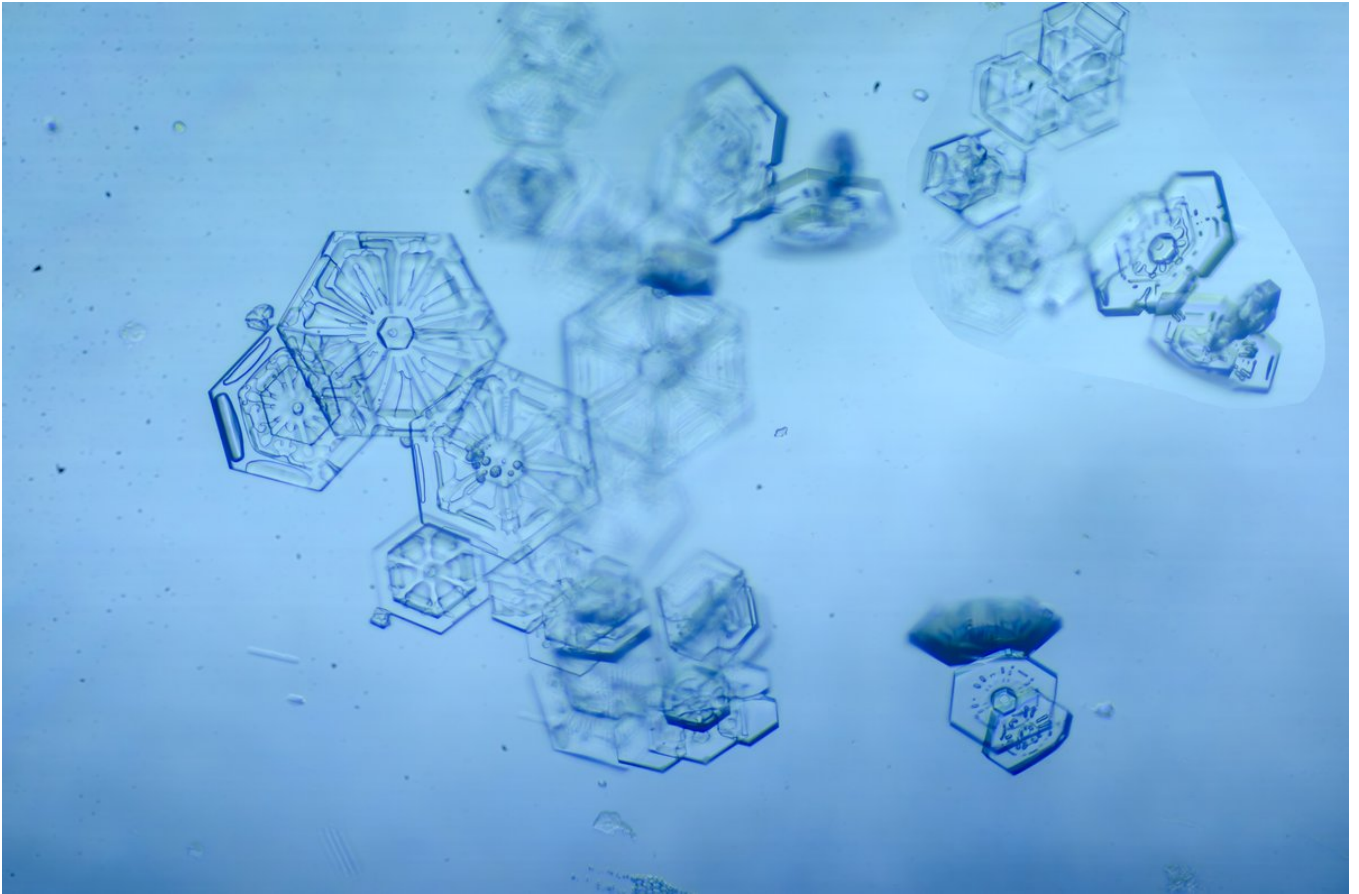
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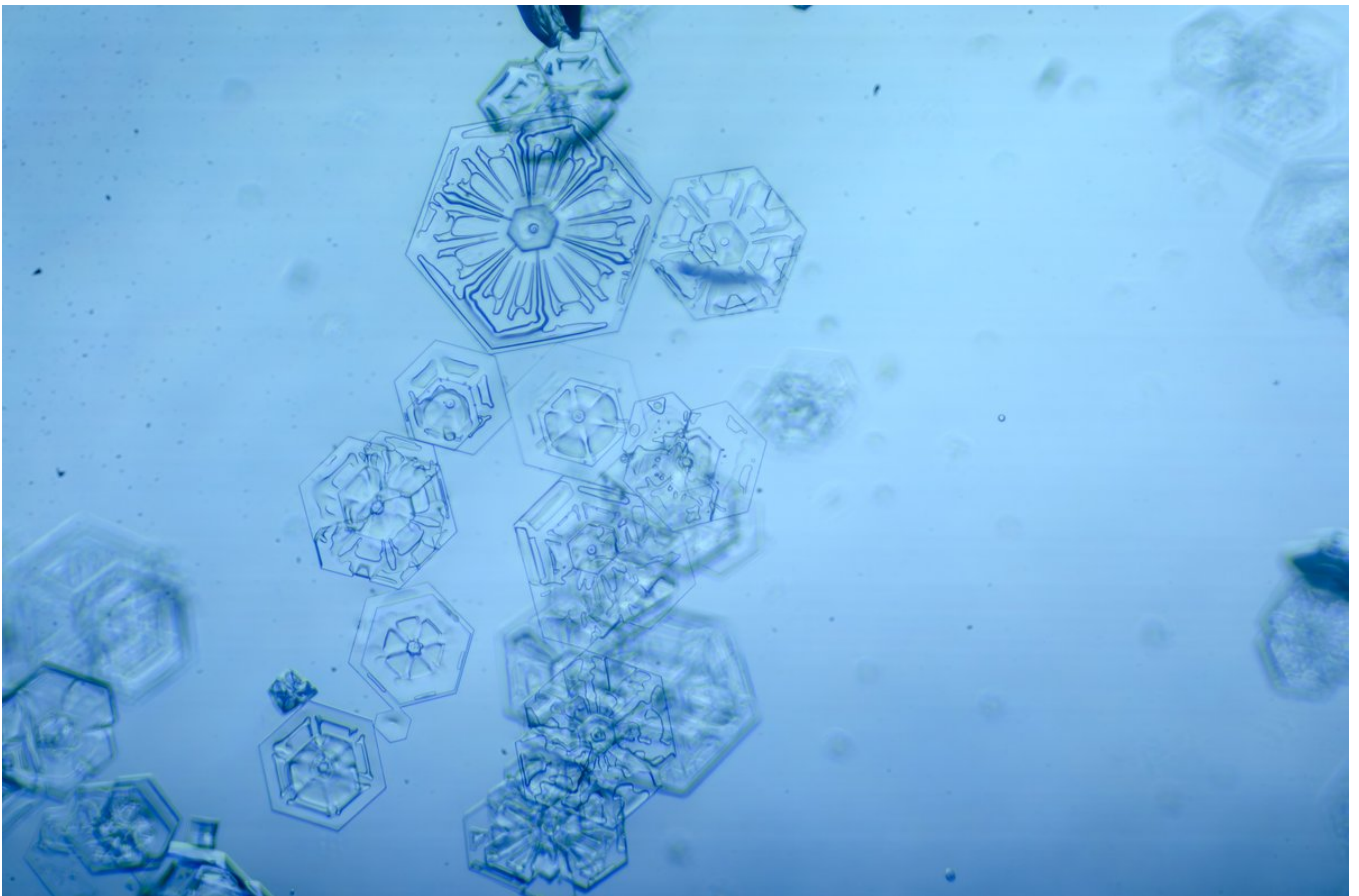
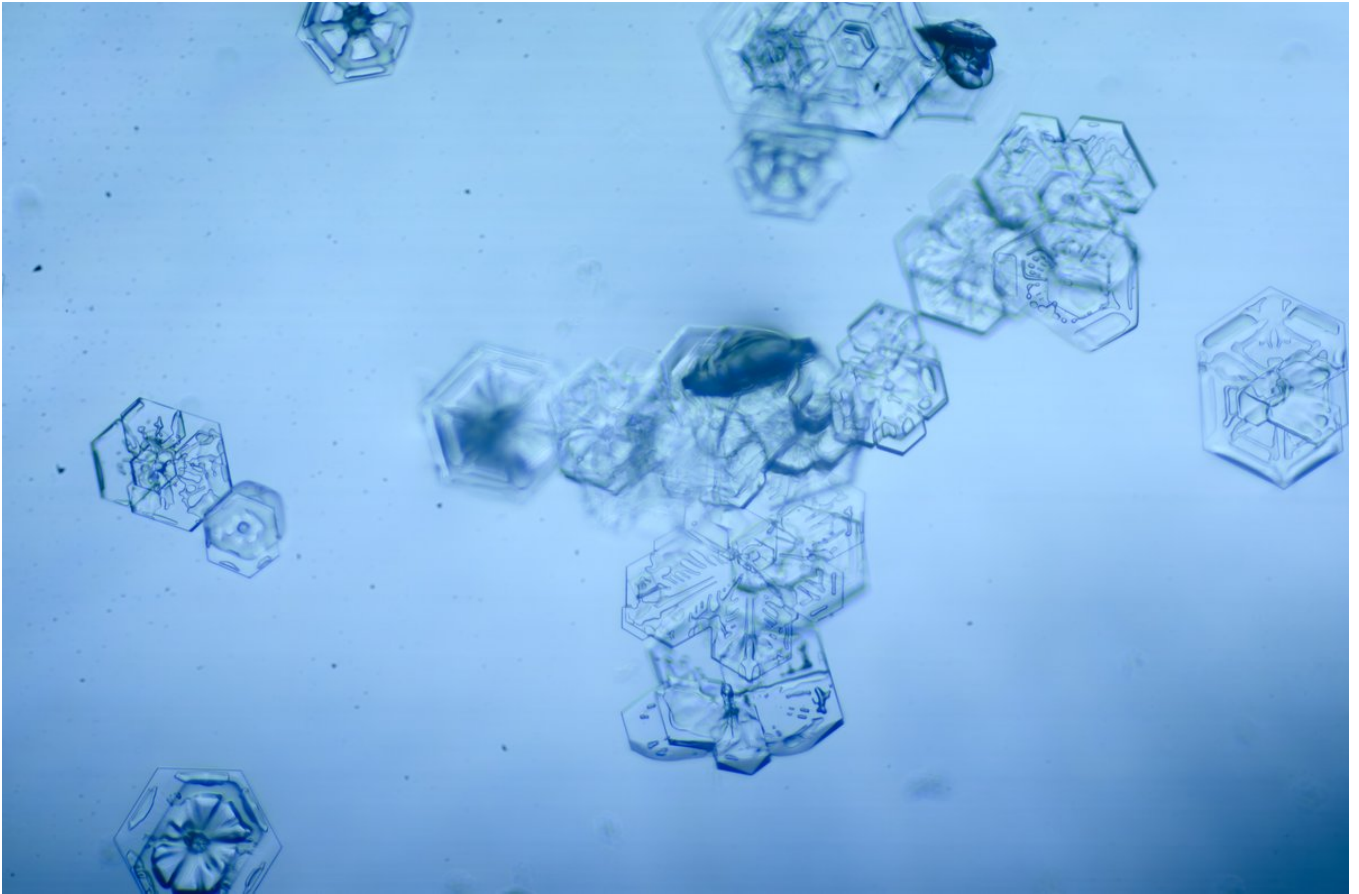




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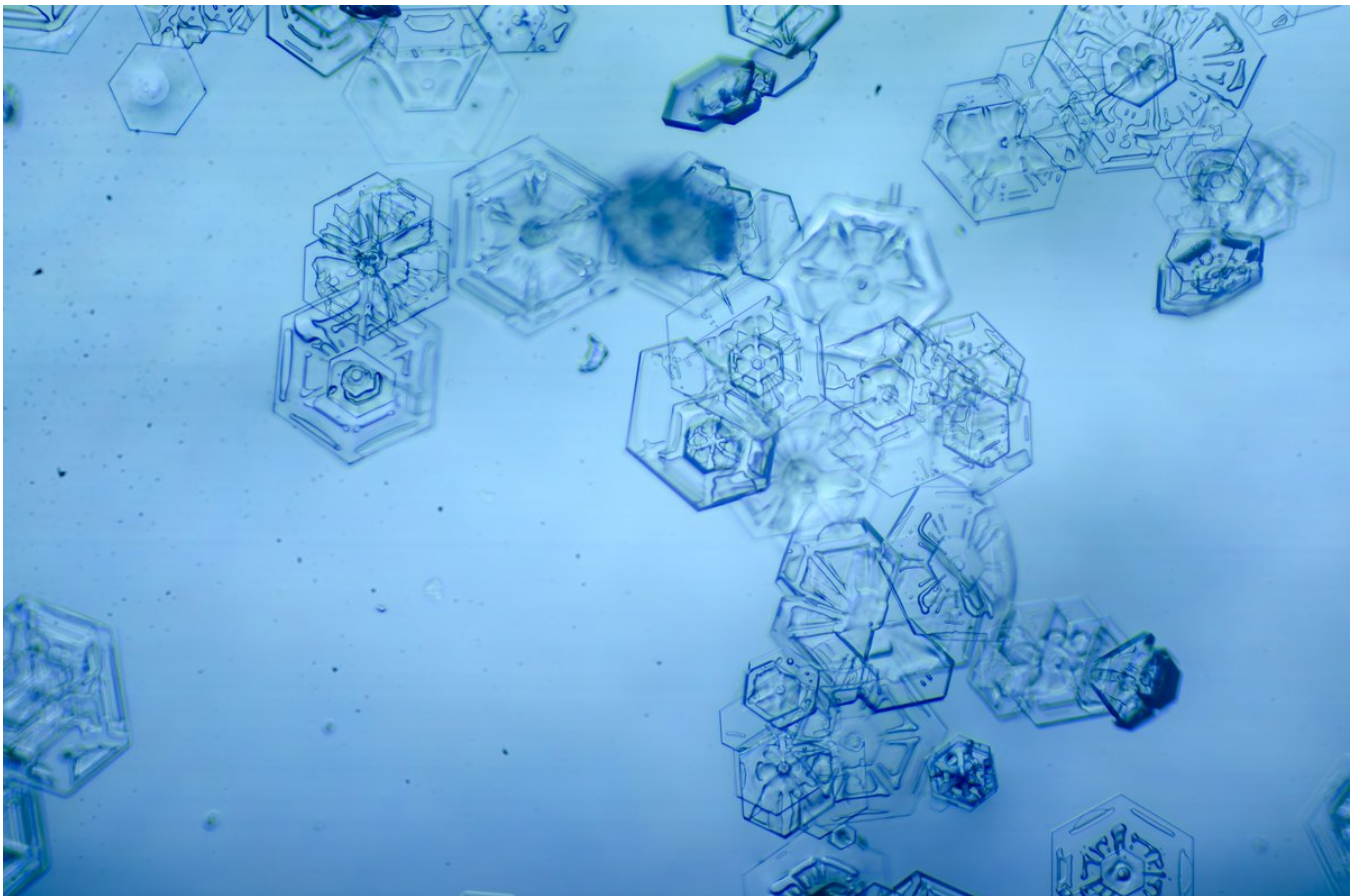
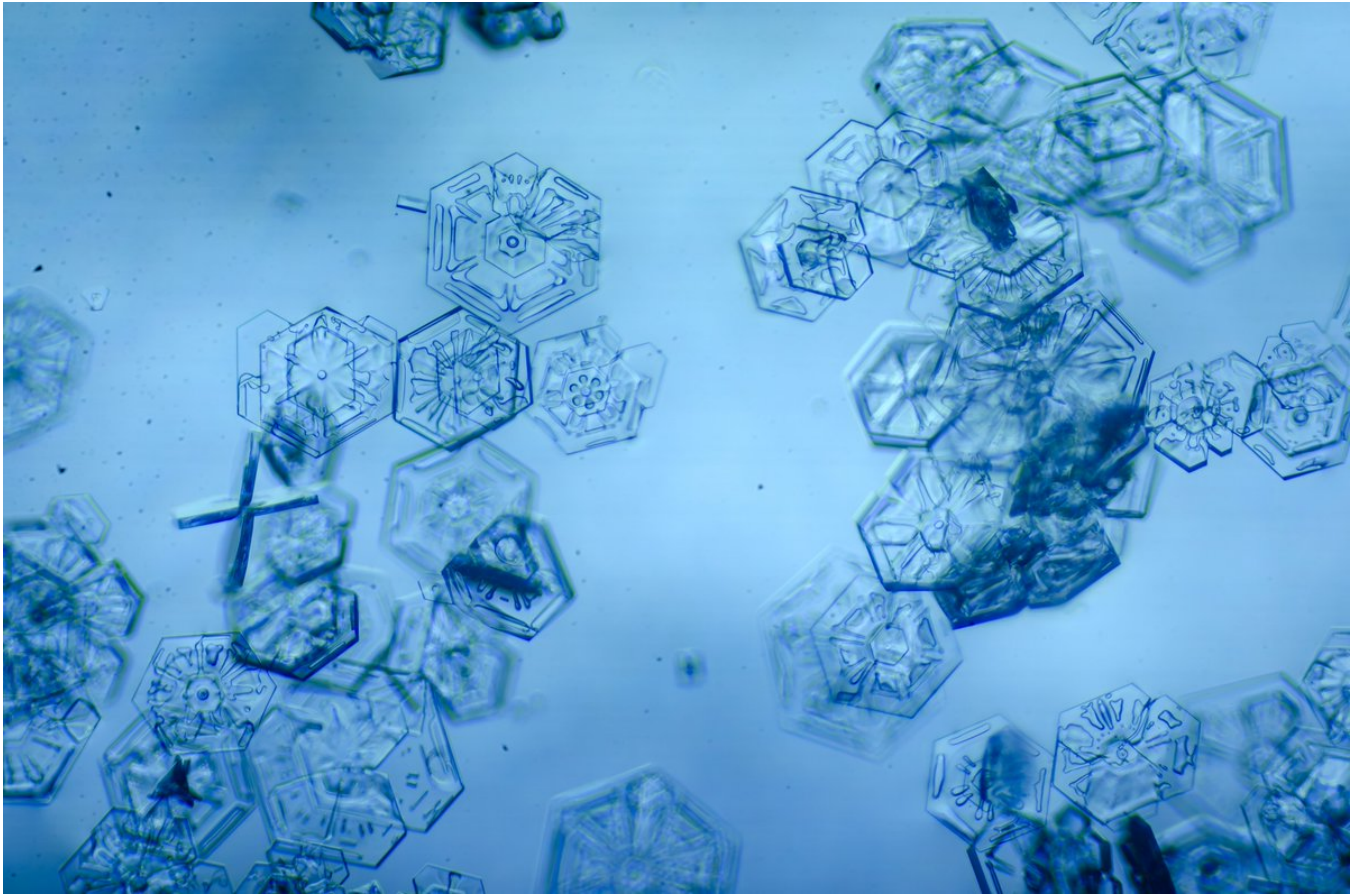
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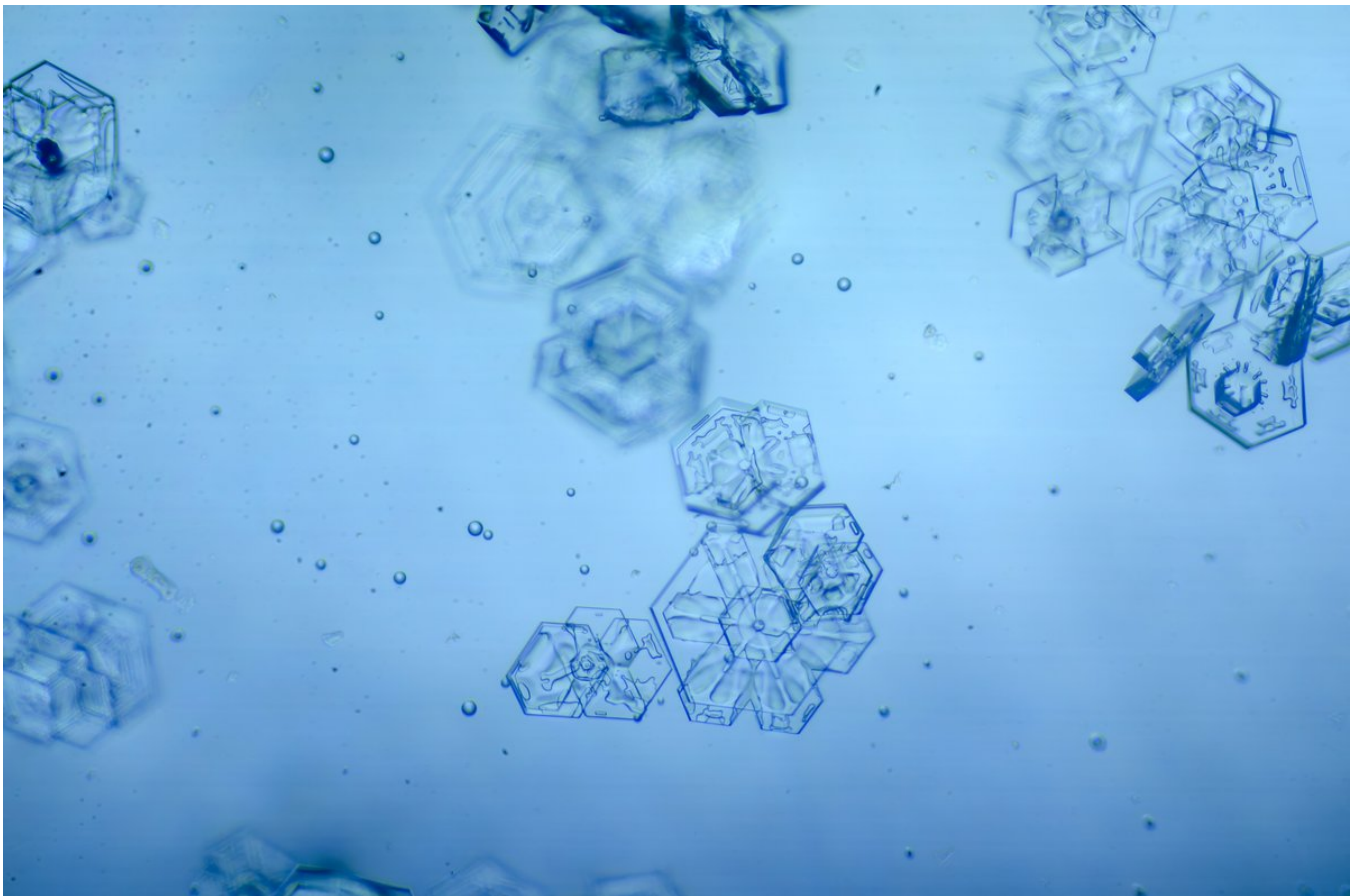
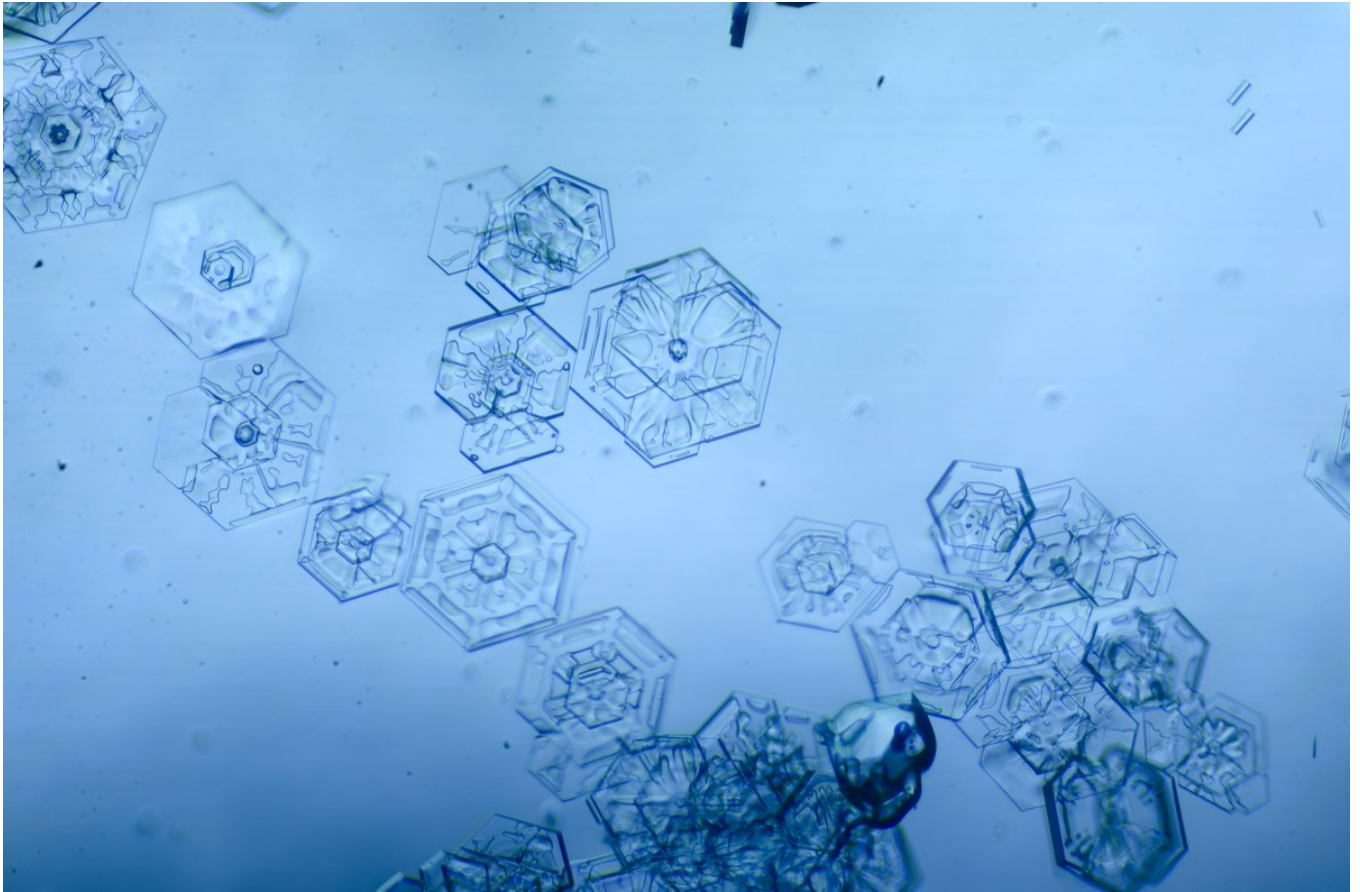




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